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香港理工大學

Department of
Building Environment and Energy Engineering
建築環境及能源工程學系

Development of new water spray strategies for improving the energy performance of indirect evaporative cooling systems

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22 July 2024

中文要点｜报告主题

新型喷淋策略开发及其对间接蒸发冷却系统能效的提升

Outline

1. **Research background and literature review**
2. **Research gaps and objectives**
3. **Development of FDM model for spray distribution optimization of IEC**
4. **Development of CFD model for IEC considering water film formation with spray parameter analysis**
5. **Performance assessment and optimization of water spray strategy for IEC based on ANN modeling and GA**
6. **Performance analysis and nozzle setting optimization of IEC for energy recovery in data centers**
7. **Conclusions and recommendations for future work**

中文要点｜汇报提纲

研究背景与目标 → FDM 喷淋分布建模 → CFD 水膜分析 → ANN–GA 喷淋优化 → 数据中心应用 → 结论与展望

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中文要点｜研究背景与文献综述

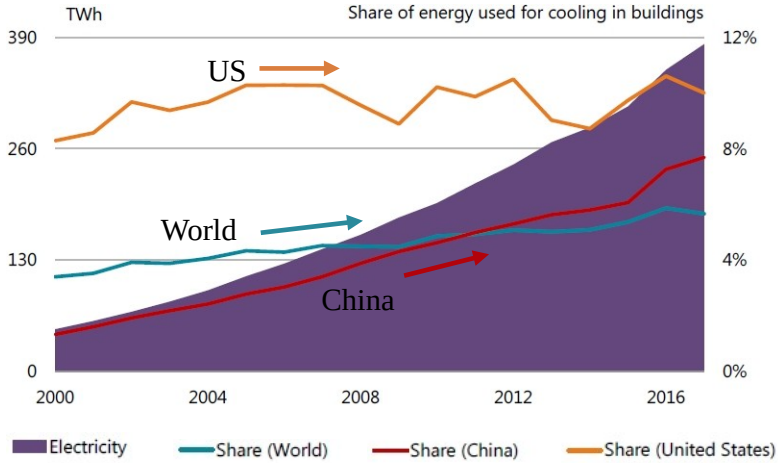
从建筑制冷能耗增长出发，介绍 IEC 技术的发展、水膜润湿性研究及现有模型的不足。

1. Research background and literature review

1.1 Cooling energy use in buildings

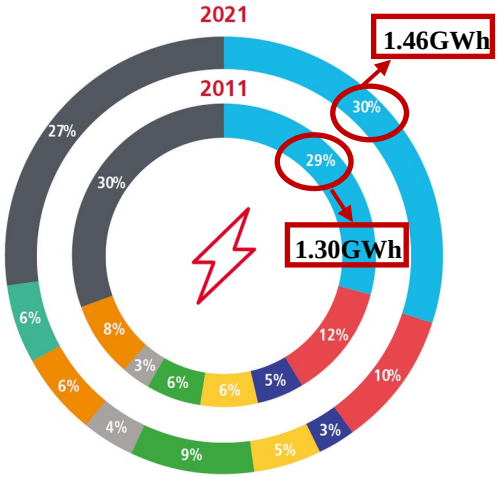
- **China:** cooling energy use has been growing fast in past 10 years, accounting for 1/5 of global cooling energy use.

- **Hong Kong:** the electricity end use for A/C increased by 29% in past 10 years and is about 30% of the total electricity use in building sector.



Electricity use for space cooling, 2000-2017

Source: International Energy Agency (IEA). The Future of Cooling in China. 2019



Electricity Consumption by End-use in HK, 2011-2021 *Source: EMSD. Hong Kong Energy End-use Data. 2023*

Seek for environmental friendly technology for cooling

中文要点｜建筑制冷能耗

中国制冷能耗约占全球的五分之一；香港过去十年空调用电增长 **29%** 。发展高效、环保的制冷技术十分必要。

1. Research background and literature review

1.2 Development of indirect evaporative cooling (IEC)

- The IEC technology is a green and energy-saving technology that uses water to evaporate and absorb heat for cooling.

  - **Conventional IEC:** for sensible cooling-dominant buildings under dry climates.

  - **Updated IEC:** for energy recovery in hot and humid regions.

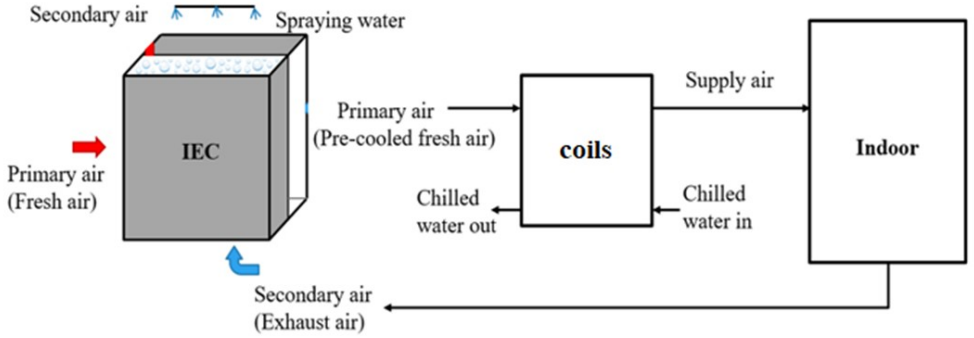


Fig.1.2.2 System diagram of an IEC for energy recovery

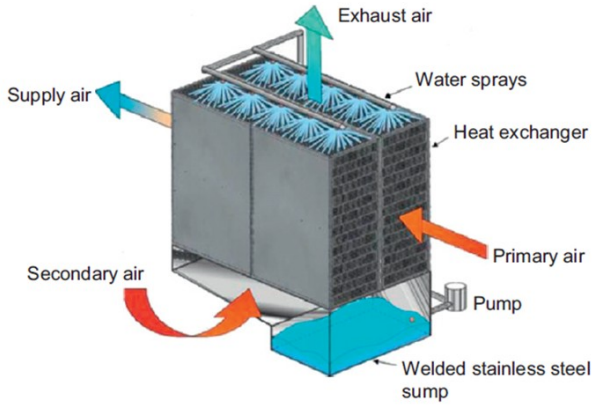


Fig.1.2.1 Schematic diagram of a traditional IEC

- **Key point:** the process of evaporation of water into air.

Enhanced !

Source: A statistical modeling approach on the performance prediction of indirect evaporative cooling energy recovery systems. 2019

中文要点｜间接蒸发冷却技术

IEC 利用水蒸发吸热实现降温；应用由干燥地区的显热冷却，拓展至湿热地区的能量回收。

Appendix

➢ Working principle

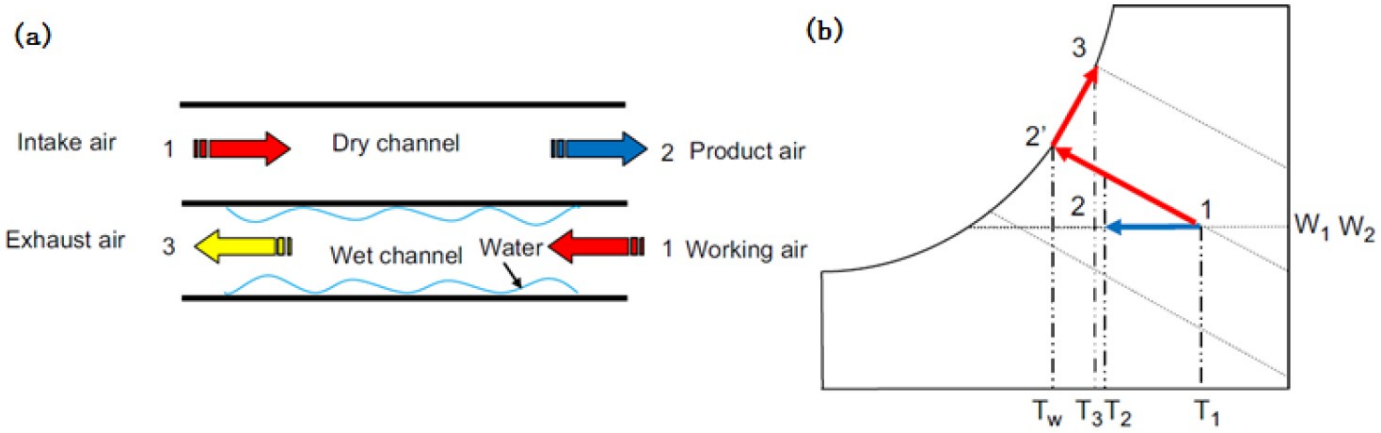


Fig. 8.1 (a) schematics and (b) psychrometric process of traditional IEC

- When IEC is applied in hot and humid areas, the **primary air** humidity is high and the wet-bulb temperature of the **secondary air (exhaust air)** in the air-conditioned space is low, resulting in condensation where the temperature of the plate surface is lower than the dew-point temperature of the primary air.

中文要点｜工作原理

湿热地区的一次空气湿度较高。当板面温度低于其露点温度时，一次空气侧发生冷凝，可同时回收显热与潜热。

1. Research background and literature review

1.3 Research on wettability of secondary air channel

- Water supply
  - Low water supply → Low wettability and heat & mass transfer
  - High water supply → Higher film thickness and thermal resistance

- Plate structure

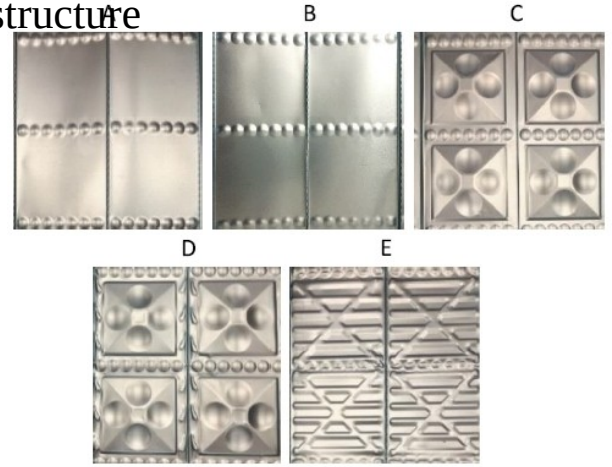


Fig.1.3.1 Plate with different pitch or protrusion geometry

- Air flow direction

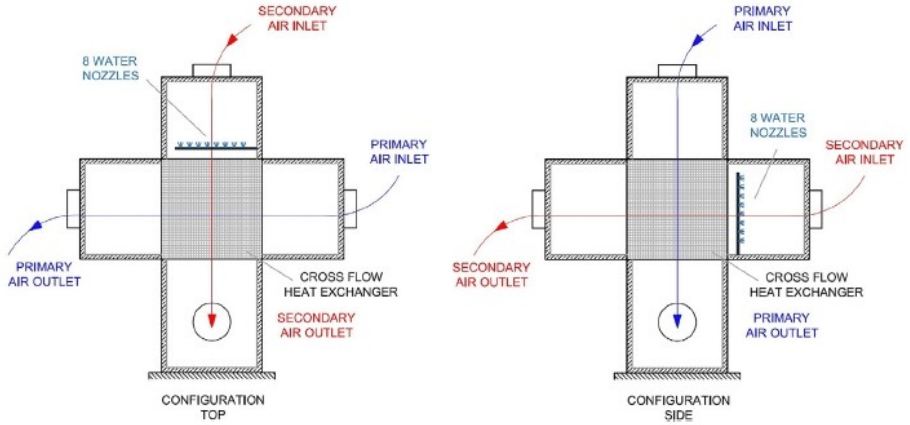


Fig.1.3.2 Scheme of Top and Side configuration

Source: Effect of plates coating on performance of an indirect evaporative cooling system. 2019;
Effect of heat exchanger plates geometry on performance of an indirect evaporative cooling system, 2020

中文要点｜湿通道润湿性

供水不足导致干区和传热传质性能下降；供水过多则使水膜增厚、热阻增大。应协同优化板面结构与供水方式。

1. Research background and literature review

- Water spray nozzle
  - Arrangement

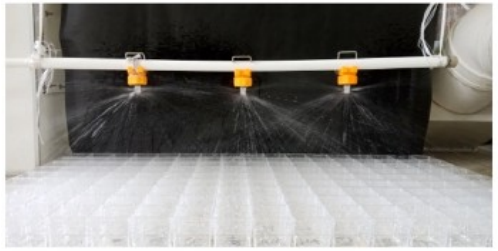


Fig.1.3.3 The photos of water distribution effect[1]

  - Types



Fig.1.3.4 Five types of commonly used spray nozzle[1]

  - Distributions

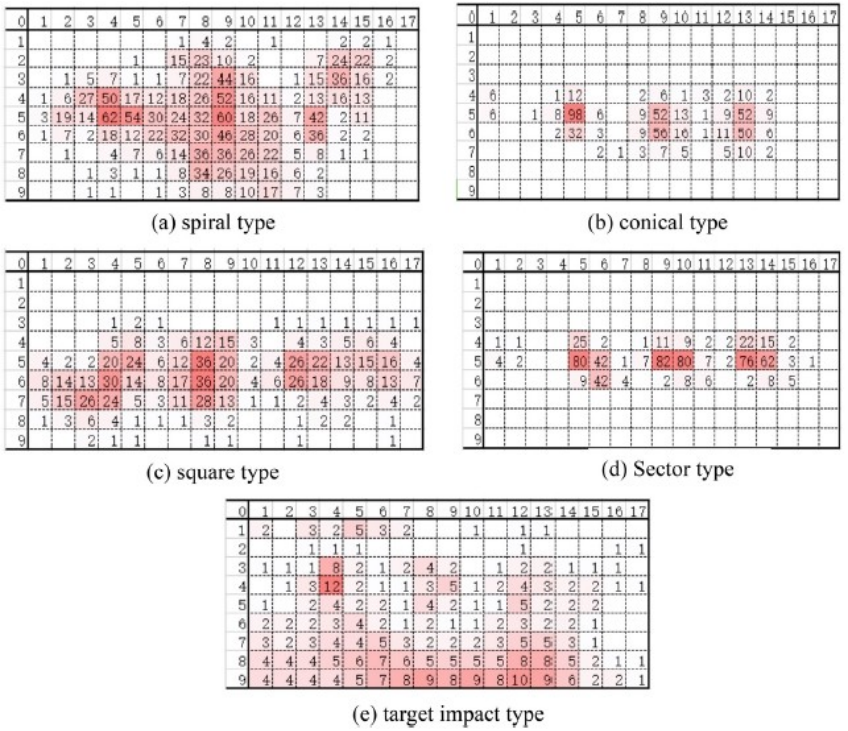


Fig.1.3.5 Water distribution volume of spraying nozzles(220 L/h) [1]

[1] Sun T, Huang X, Chen Y, Zhang H. Experimental investigation of water spraying in an indirect evaporative cooler from nozzle type and spray strategy perspectives.

中文要点｜喷嘴与布水特性

喷嘴类型、布置方式及喷淋分布共同决定板面的布水效果，是改善润湿性的重要设计因素。

1. Research background and literature review

1.4 Research on previous model

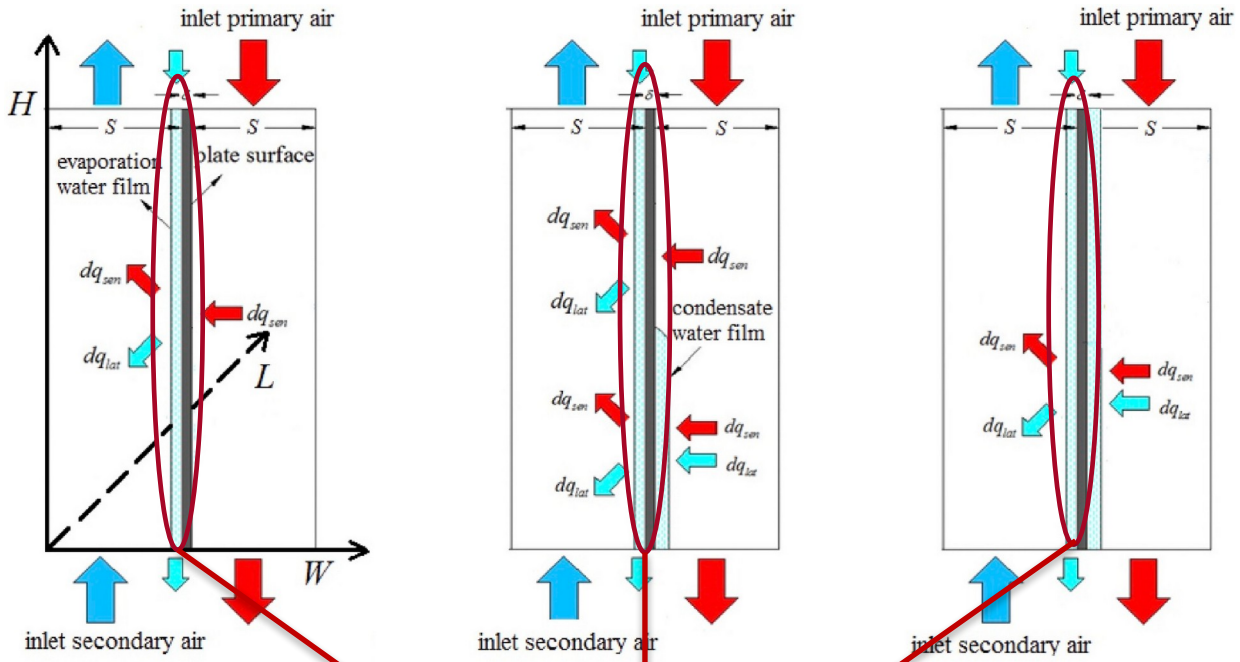


Fig.1.3.5 Previous model of the IEC

- **Assumption:** the thin water film entirely and evenly covered the surface of the wet channel.

Ideal

中文要点｜现有模型的局限

“水膜完整且均匀覆盖湿通道”的理想假设与实际情况存在差异，需考虑不完全润湿与真实喷淋分布。

1. Research background and literature review

1.5 IEC as the recovery unit

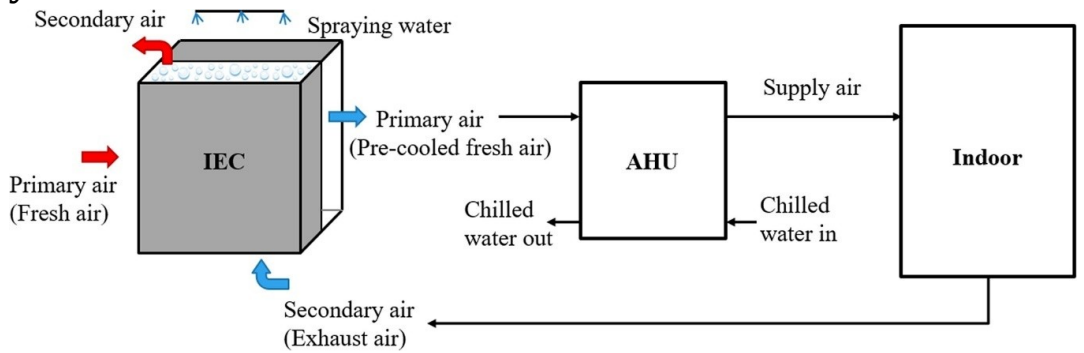
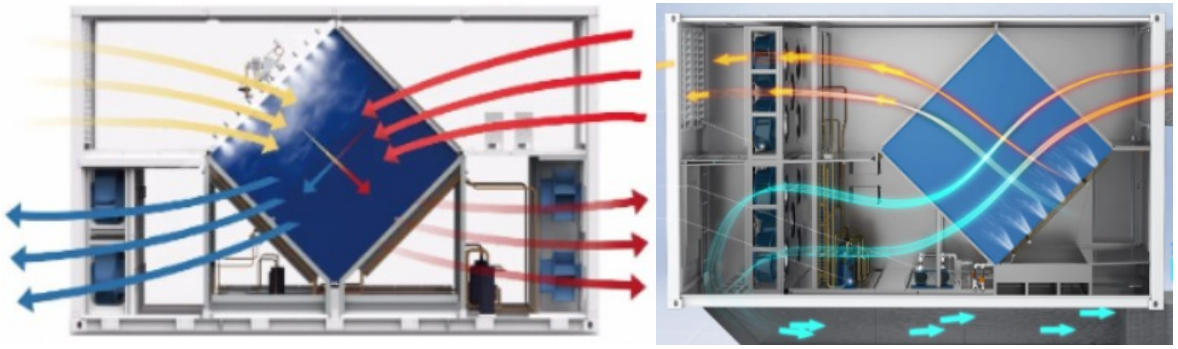


Fig.1.5.1 Schematics diagram of IEC followed by AHU

- Various nozzle settings for diamond-shaped IEC



A more ***standardized nozzle arrangement guideline*** has not yet emerged.

中文要点｜IEC 能量回收应用

IEC 可与空气处理机组（AHU）结合使用，但菱形 IEC 仍缺少标准化的喷嘴布置指导方法。

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中文要点｜研究不足与目标

围绕实际水膜分布、喷淋参数耦合和工程优化需求，建立从机理模型到智能预测的研究体系。

2. Research gaps and objectives

2.1 Research gaps

1. Most of the existing analysis about water spray of IEC are based on experimental data, and few mathematical models can **quantitatively simulate** the various parameters.
2. The existing heat and mass transfer model of IEC has not fully discussed on the **wetting ratio**, and the ideal assumption needs to be corrected.
3. The current 2D model is not enough to accurately simulate the state of the **water film distribution**, and to deal with more complex hydraulic and thermal properties.
4. The **optimal nozzle arrangement** and **parameter setting** to maintain the best wetted situation of the IEC wet channel have not been identified.
5. There is a need for **straightforward method** to predict the IEC performance for use in reality and optimize spray design parameters in large scale projects.
6. The **diamond-shaped IEC** used as heat recovery units combined with AHU in DCs require targeted water spray system settings.

中文要点｜六项研究不足

现有研究在定量喷淋建模、真实润湿率、三维水膜模拟、最优喷嘴设计、高效预测及数据中心适配方面仍有不足。

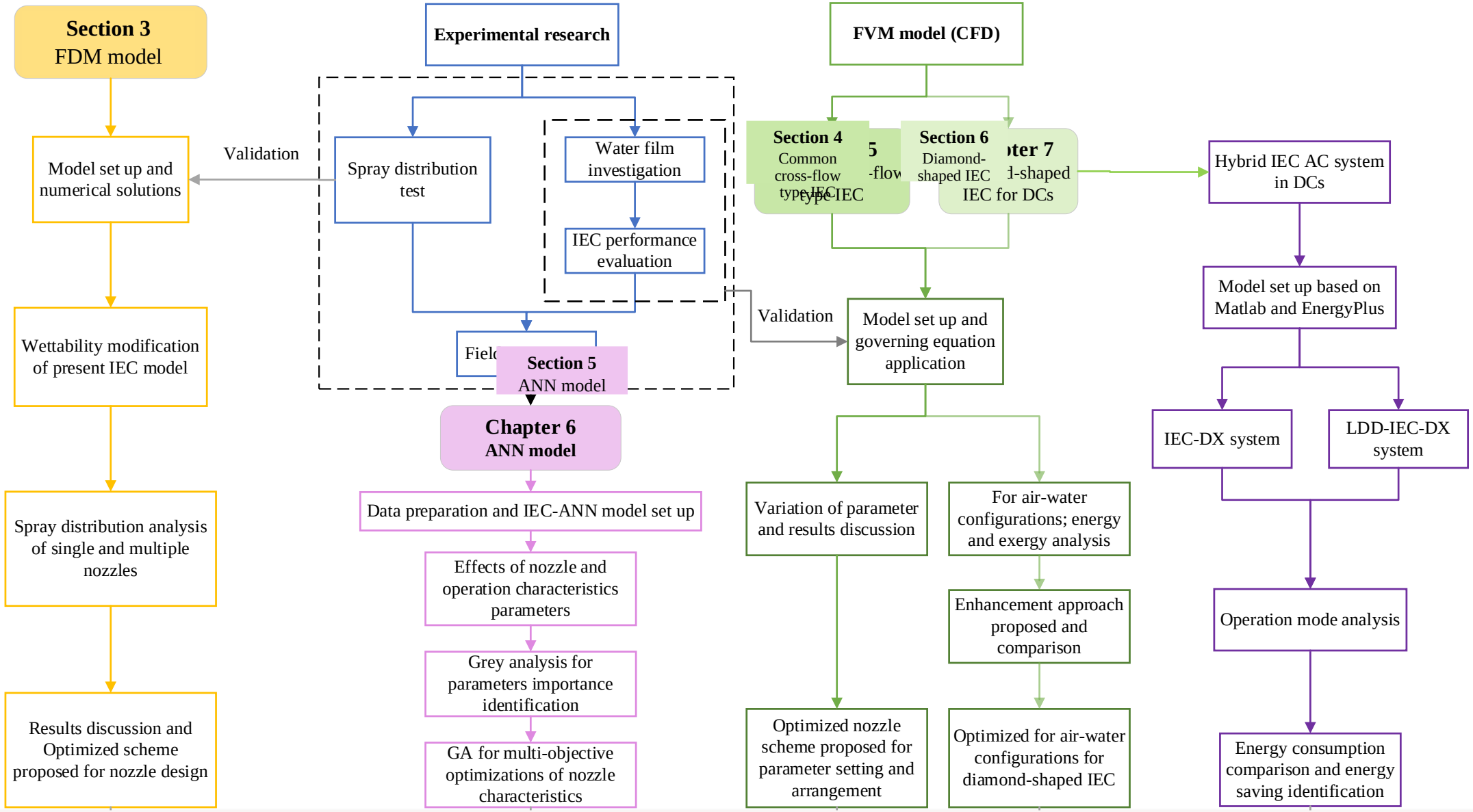
2. Research gaps and objectives

2.2 Research objectives

1. To propose a **FDM model** for predicting the **water spray distribution** of the nozzles in IEC.
2. To establish a **3D CFD model** to identify the formation and flow state of the water membrane covering the plate surface.
3. To set up the **test bench** with the complete IEC system that can investigate the **practical wetted sections** in single channel pair for model verification.
4. To conduct the **parametric analysis** and determine the **optimal water spraying strategies** for IEC system.
5. To develop high-efficiency prediction model based on **ML** to accurately predict the performance of IEC system combined with **genetic algorithm** for multi-objective **optimization**.
6. To establish and verified a **CFD model** based on the **diamond-shaped IEC** to explore the optimal nozzle configurations and performances, as well as the enhancement strategies for IEC in DCs

中文要点｜研究目标

建立并验证 FDM、三维 CFD 和 ANN 模型，结合实验与遗传算法，优化常规及菱形 IEC 的喷淋策略。



中文要点｜研究技术路线

以 FDM 研究喷淋分布，以 CFD 分析水膜形成，以 ANN 提高预测效率，并拓展至菱形 IEC 的应用优化。

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中文要点｜第三部分： FDM 模型

建立喷淋分布模型，通过实验验证与多喷嘴方案对比，优化板面润湿性及 IEC 能量回收性能。

3. Development of FDM model for nozzle optimization of IEC

3.1 Mathematical model

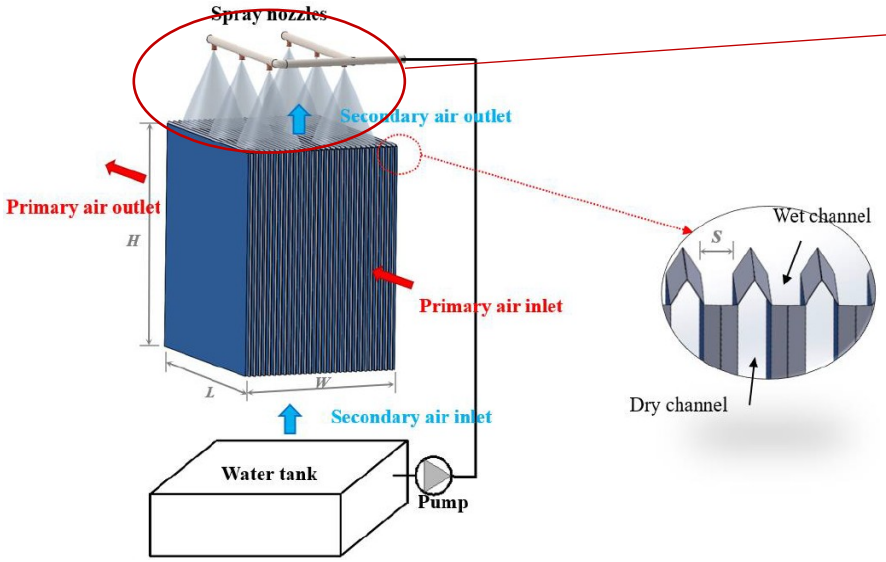


Fig. 3.1.1 Configuration of an IEC system

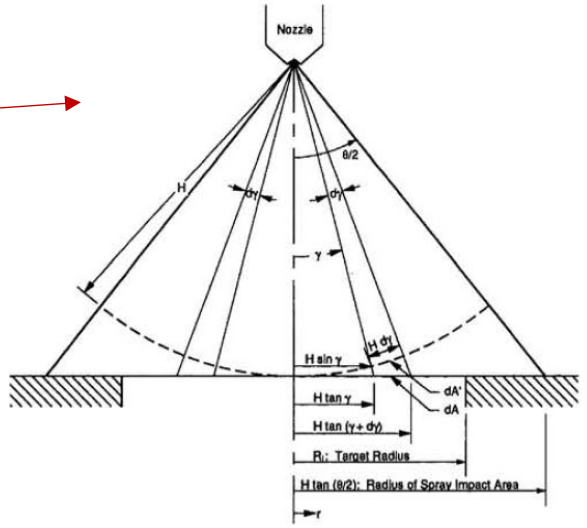


Fig. 3.1.2 Geometry used in volumetric flux distribution model [1]

Assumption：

The total spray flow rate, Q, is uniformly distributed over any spherical surface centered at the spray orifice.

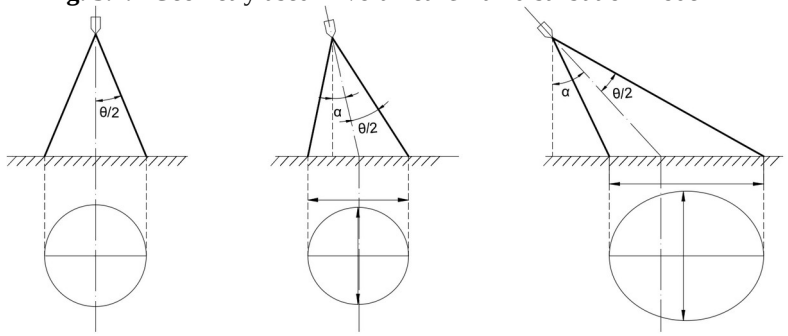


Fig. 3.1.3 Three scenarios of spray impact area

[1] Mudawar I, Estes KA. Optimizing and predicting CHF in spray cooling of a square surface. J Heat Transfer 1996;118:672–9. https://doi.org/10.1115/1.2822685.

中文要点｜数学模型与假设

以喷口为球心，假设总喷淋流量 Q 在任一球面上均匀分布，据此计算不同区域的喷淋通量。

3. Development of numerical model for nozzle optimization of IEC

3.2 Model equations

- Boundary condition:

$$x^2 + y^2 = \left(H \times \tan\frac{\theta}{2}\right)^2$$

- Imaginary spherical surface:

$$\frac{(x-x_0)^2}{a^2} + \frac{(y-y_0)^2}{y^2} = 1$$

$$R = \sqrt{x^2 + y^2 + H^2}$$

$$cos\alpha = \frac{H}{R}$$

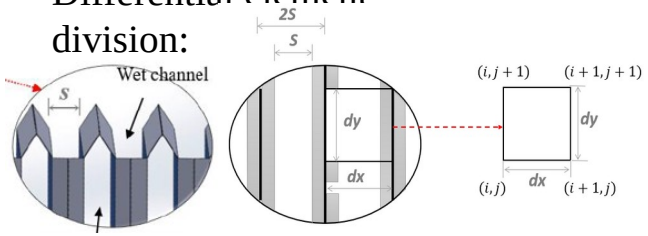
$$dA = 2\pi R^2 \times \frac{sin\alpha}{cos\alpha} d\alpha$$

- Volume flux of spray surface area:

$$A = 2\pi R^2 \times \frac{\left[1 - \cos\left(\frac{\theta}{2}\right)\right]}{cos\alpha}$$

$$Q'' = \frac{Q}{2\pi R^2 \times \left[1 - \cos\left(\frac{\theta}{2}\right)\right]} \times cos\alpha$$

- Differential element division:



$$Q_{(i,j)} = \int_{x_i}^{x_{i+1}} \int_{y_i}^{y_{i+1}} \frac{Q \times H}{2\pi\left(\sqrt{x^2 + y^2 + H^2}\right)^3 \times \left[1 - \cos\left(\frac{\theta}{2}\right)\right]} dxa$$

- Grid independence:

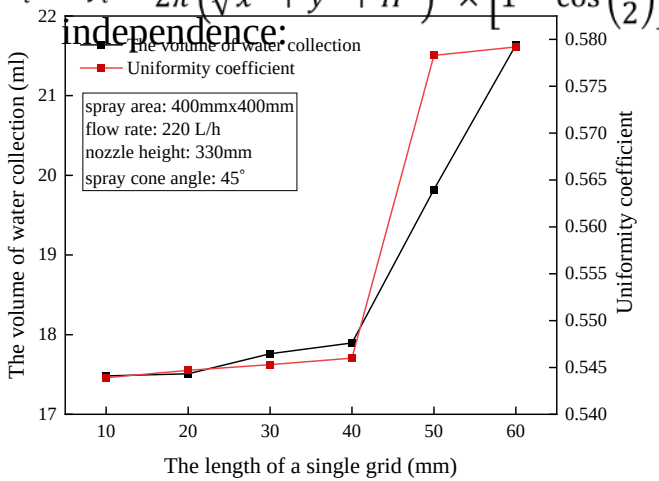


Fig.3.2.1 Proof of grid independence

- Simulation process (MATLAB)

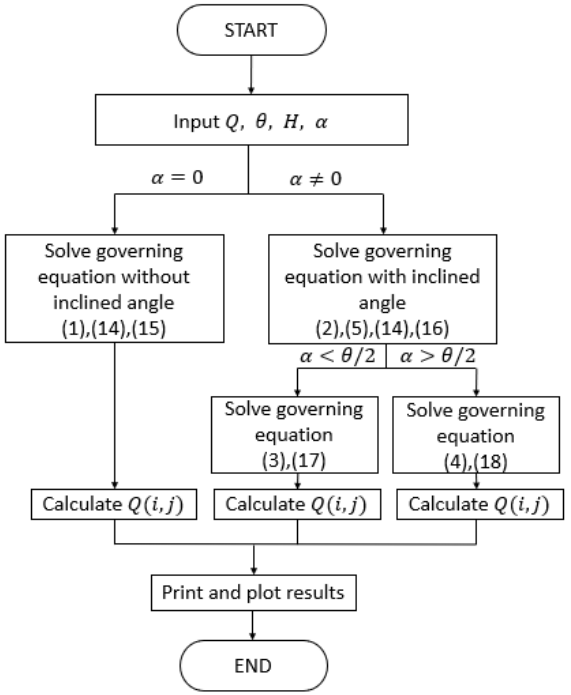


Fig. 3.2.2 Flow chart of simulation procedure

中文要点｜模型方程与求解

通过球面几何与微元划分计算局部体积通量，并在 MATLAB 中完成求解与网格无关性验证。

3. Development of numerical model for nozzle optimization of IEC with experimental validation

3.3 Experiment validation

- Condition:

Q=220 L/h,

Spray cone angle 45°,

Nozzle height 330 mm without inclined angle,

Grid number is 10 × 10,

Grid size i

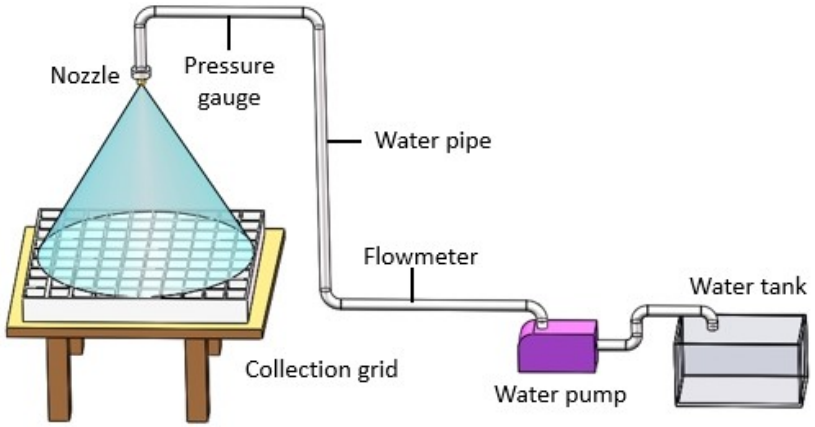


Fig. 3.3.1 Experiment bench

- Validation result:

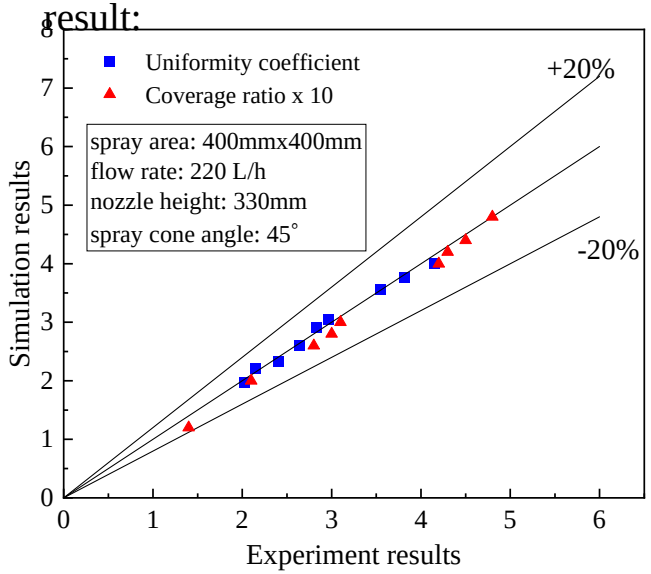


Fig. 3.3.2 Validation results

The maximum discrepancy is acceptable.

中文要点｜喷淋分布模型验证

搭建实验台，对比实验与模拟结果；模型最大偏差在可接受范围内，可用于后续喷嘴优化分析。

Appendix

- **Instrument selection**

Table 8.1 Solid nozzle selection

Nozzle	Orifice diameter (mm)	Volumetric flow rate	Pressure	Spray angle
1	2.4	1.9-8.4	0.4-10	45-50
2	3.0	2.9-13	0.4-10	60-70
3	3.2	6.7-16.2	0.4-10	70-75

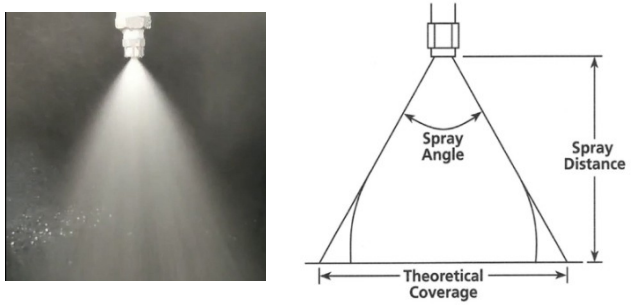


Fig. 3.3.1 Photos of the nozzles

Table 8.2 Instrument parameters list

Instrument	Parameter	Range	Accuracy
Flowmeter	Water flow rate	1.9-8.4	0.4-10
Pressure gage	Water pressure	2.9-13	0.4-10
Measuring cylinder	Water volume	6.7-16.2	0.4-10



Fig. 8.3 Photos of test instrument

中文要点｜实验仪器与喷嘴选型

根据孔径、流量、压力和喷雾锥角选择喷嘴，并通过流量计、压力表与量筒开展喷淋测量。

3. Development of numerical model for nozzle optimization of IEC with experimental validation

3.4 Performance indexes

➢ Water distribution

Uniformity coefficient: $\xi = \frac{1}{\bar{x}}\left(\sqrt{\frac{\sum_i^N x_i - \bar{x}}{N}}\right)$

Coverage ratio: $\gamma = \frac{S_w}{S}$

➢ Water membrane

Reynolds number: $Re_w = \frac{m_w}{L_f \mu_w}$

Velocity: $v_w = 1.5 \times \left(\frac{\mu_w g}{48}\right)^{\frac{1}{3}} (Re_w)^{\frac{2}{3}}$

Spraying density: $\Gamma = \frac{\dot{m}_w}{(n+1)L_f}$

Water membrane thickness: $\delta_w = \left(\frac{3\mu_w \Gamma}{}\right)^{\frac{1}{3}}$

➢ IEC performance

Pressure drop: $\triangle P = \frac{f_{Re} L \rho u^2}{2Red_e}$

Fan consumption:

$W_{fan} = \frac{Q \triangle P}{3600 \times 1000 \times \eta_0 \times \eta_1} \times K$

Pump consumption:

$W_{pump} = m_w g(h_{gravity} + h_{nozzles} + h_{valves}) \times$

The coefficient of performance:

$COP = \frac{Q_p}{W_{fan} + W_{pump}}$

Wet-bulb efficiency:

$\eta_{wb} = \frac{t_{p,in} - t_{p,out}}{t_{p,in} - t_{wb,s,in}}$

Cooling capacity:

$Q_c = m_p c_{pa}(t_{p,in} - t_{p,out})$

中文要点｜性能评价指标

采用湿球效率和制冷量等指标，评价喷淋优化对 IEC 热湿处理能力与运行性能的影响。

3. Development of numerical model for nozzle optimization of IEC with experimental validation

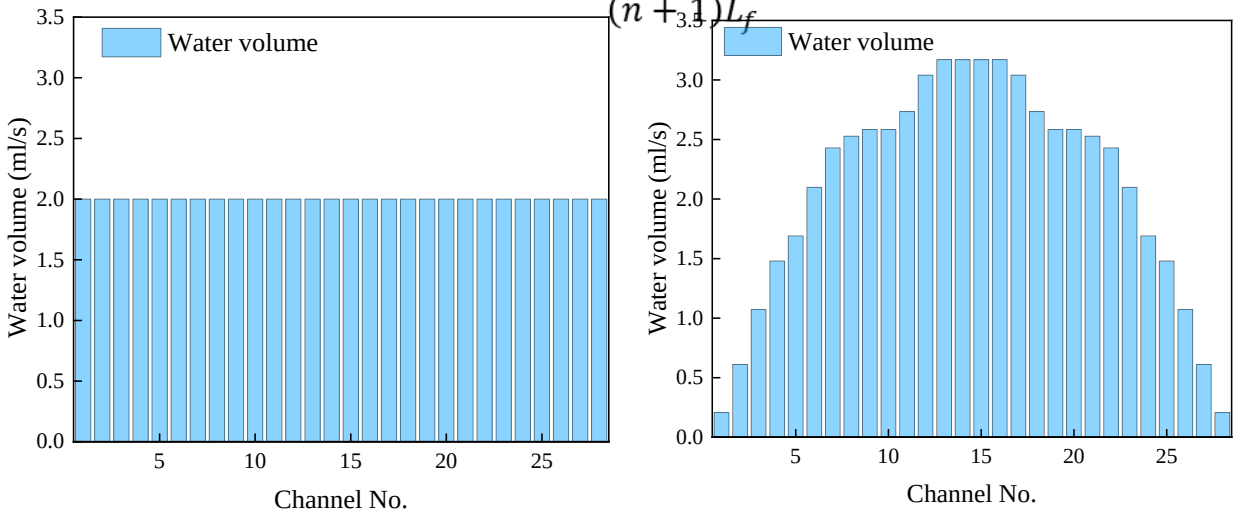
3.5 Results and discussions

3.5.1 The water density and wettability in channels

- Water density modification

Original spraying density:

$$\Gamma = \frac{\dot{m}_w}{(n+1)L_f}$$



Ideal scenario

Reality scenario

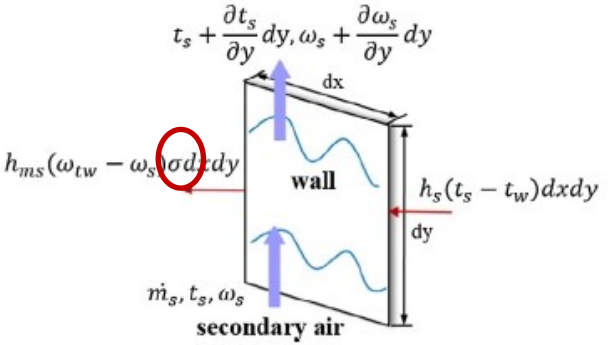


Fig. 3.5.1 Heat and mass transfer of the secondary air flow channel

中文要点｜喷淋密度与润湿修正

将真实喷淋密度和局部润湿状态纳入湿通道传热传质分析，修正理想完全润湿假设。

3. Development of numerical model for nozzle optimization of IEC with experimental validation

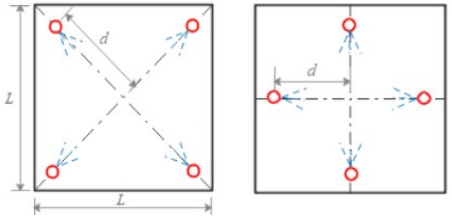
3.5 Results and discussions

3.5.2 Results of multiple nozzles

- Arrangement schemes

The size of heat exchanger: 400 mm × 400 mm

The height of the nozzle: 300 mm

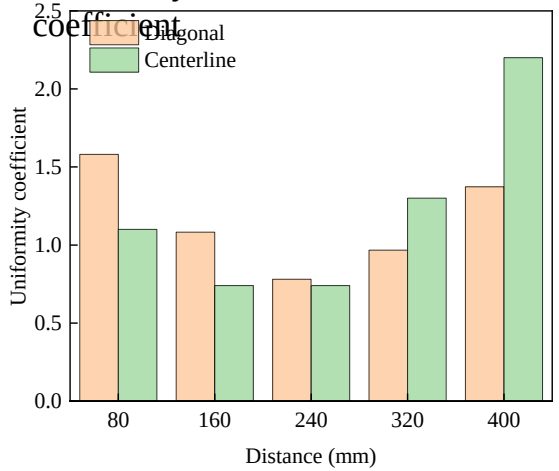


(a) Diagonal (b) Centerline

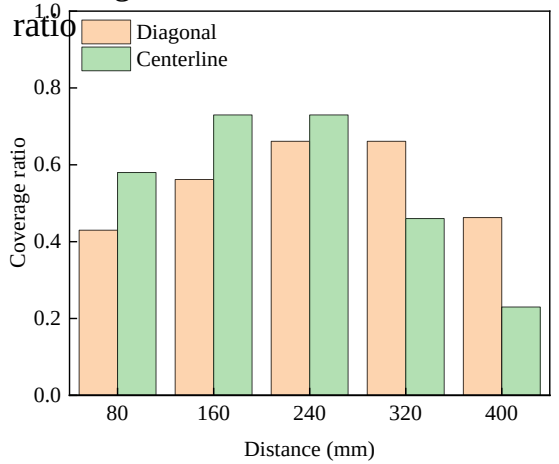
Fig. 3.5.2 Nozzle arrangement schemes

- The effect of the nozzle distance on the spray distribution

  - Uniformity coefficient



  - Coverage ratio



- Spray optimal arrangement: Centerline with 160 mm distance.
- Uniformity coefficient is 0.74 and coverage ratio is 0.72.

中文要点｜多喷嘴最优布置

本研究中，沿中心线布置、间距 **160 mm** 时效果最佳；均匀性系数为 **0.74**，覆盖率为 **0.72**。

3. Development of numerical model for nozzle optimization of IEC with experimental validation

3.5 Results and discussions

3.5.3 Comparison between original and optimized schemes

- Original arrangement scheme

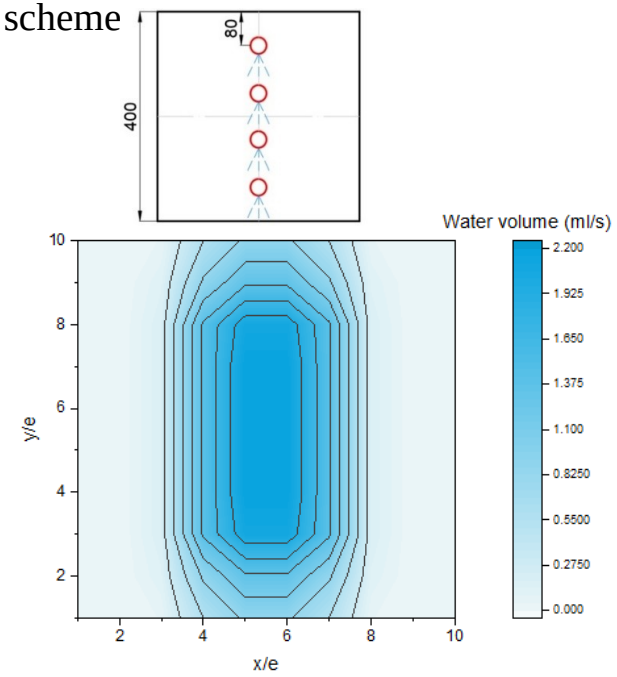


Fig. 3.5.3 Original arrangement scheme

- Wettability value: 0.48

- Optimized arrangement scheme

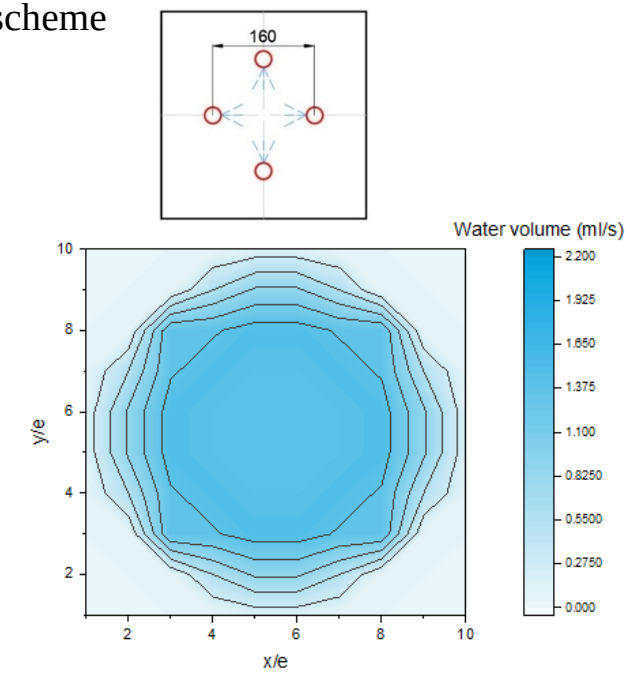


Fig. 3.5.4 Optimized arrangement scheme

- Wettability value: 0.89 **46%** ↑

中文要点｜原始与优化方案对比

优化喷嘴布置后，润湿系数由 **0.48** 提高至 **0.89**，表明合理布水可显著改善板面润湿状态。

3. Development of numerical model for nozzle optimization of IEC with experimental validation

3.5 Results and discussions

3.5.4 Performance improvement with optimized nozzle arrangement

- For dehumidification

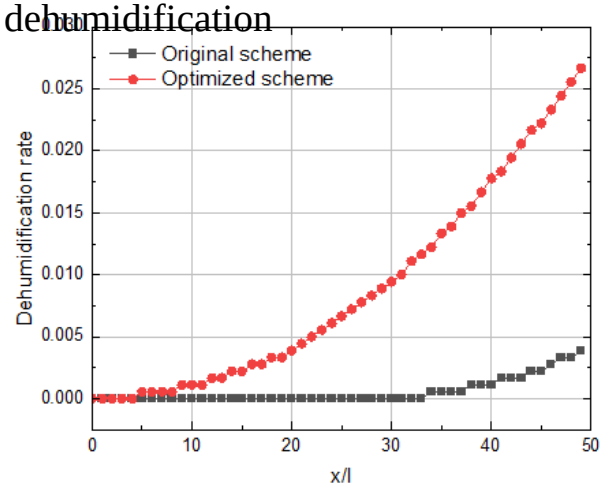


Fig. 3.5.5 Average value of the dehumidification rate

- For COP value

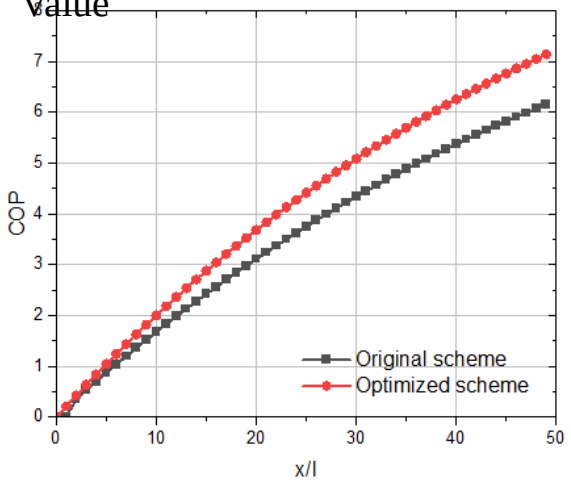


Fig. 3.5.6 Average value of the COP

- The improvement of dehumidification rate is 5.85% .
- The improvement of COP value is 16% .

中文要点｜系统性能提升

优化喷嘴布置后，除湿率提高 **5.85%**，性能系数（COP）提高 **16%**。

Outline

1. Research background and literature review
2. Research gaps and objectives
3. Development of FDM model for spray distribution optimization of IEC
4. **Development of CFD model for IEC considering water film formation with spray parameter analysis**
5. Performance assessment and optimization of water spray strategy for IEC based on ANN modeling and GA
6. Performance analysis and nozzle setting optimization of IEC for energy recovery in data centers
7. Conclusions and recommendations for future work

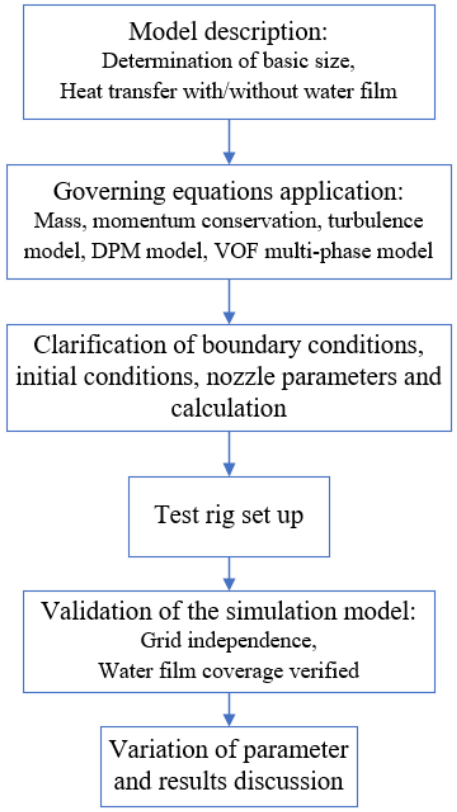
中文要点｜第四部分： CFD 水膜模型

建立考虑喷雾液滴与板面水膜的三维模型，分析喷淋参数、水膜覆盖及 IEC 性能之间的关系。

4. Development of CFD model for IEC considering water film formation with spray parameter analysis

4.1 Model description (ANSYS)

- Flowchart



- Model of single wet channel

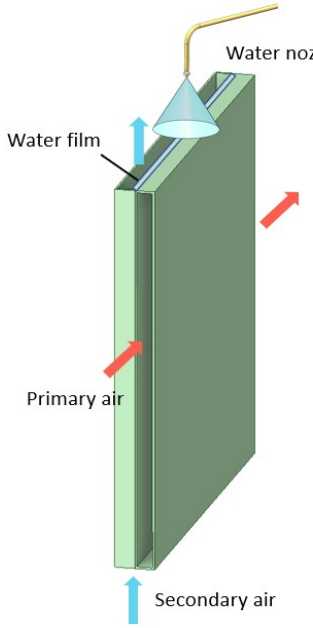


Fig.4.1.2 Configuration of an IEC system

- Different heat transfer for different coverage area

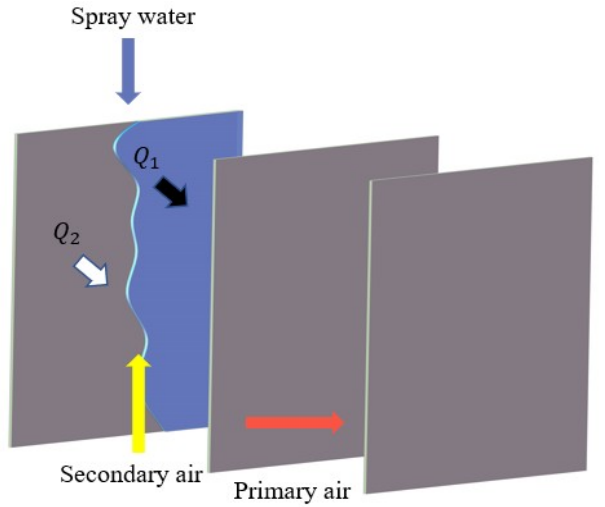


Fig.4.1.3 Geometry used in volumetric flux distribution model

Coverage factor:

中文要点｜ CFD 模型构建

针对单个湿通道建模，区分润湿与未润湿区域的传热过程，并引入覆盖系数描述实际水膜状态。

4. Development of CFD model for IEC considering water film formation with spray parameter analysis

4.2 Model equations

(1) Conservation of mass

$$\frac{\partial}{\partial t}(\rho) + \nabla \cdot (\rho u) = 0$$

(2) Conservation of momentum

$$\frac{\partial}{\partial t}(\rho E) + \nabla \cdot (\rho u^2)$$

(3) Mass transfer

$$= -\nabla P + \nabla \cdot \left(\mu(\nabla u + \nabla u^T)\right) + \rho g + F$$

$$h_{ms}\left(\omega_{t_w} - \omega_s\right)\sigma \cdot dxdy = \dot{m}_s \frac{\partial \omega_s}{\partial y} \cdot dy$$

(4) VOF multi-phase model

$$\frac{\partial \omega_e}{\partial y} = \dot{m}_s \frac{\partial \omega_s}{\partial y}$$

$$\rho = \alpha_1 \rho_1 + \alpha_g + \rho_g$$

(5) Discrete phase model (DPM)

$$\frac{d(\overrightarrow{X_g})}{dt} = \overrightarrow{V_g}$$

$$\frac{d(m_d \overrightarrow{F_d})}{dt} = \overrightarrow{F_D} + \overrightarrow{F_g}$$

$$\overrightarrow{F_d} = -\frac{\pi}{8} C_D \rho_a D_d^2 \vec{V_r} \left|\vec{V_r}\right|$$

(6) Turbulence model

Applied $k-\varepsilon$ turbulence model

(7) Calculation method

$$Re_s = 4\bar{\omega}\delta/\mu$$

EWF coupled DPM model

The SIMPLE and PRESTO! pressure

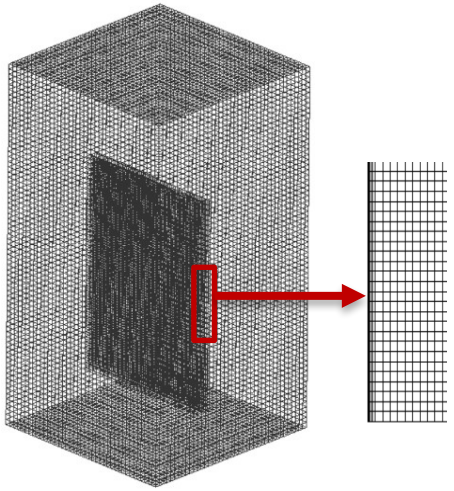


Fig.4.2.1 Sketch of meshing in the model

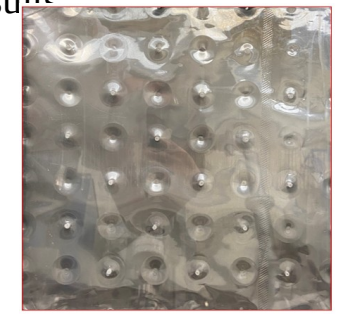
中文要点｜控制方程与耦合求解

模型考虑质量、动量与传质过程，并采用欧拉壁膜（EWF）与离散相模型（DPM）耦合描述喷雾及水膜。

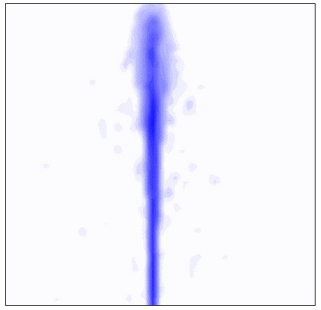
4. Development of CFD model for IEC considering water film formation with spray parameter analysis

4.3 Model validation

- Illustration of the experimental and numerical results



(a) Experimental plate



(b) Numerical



(c) Experimental result in ImageJ

- The **ICEM** software in ANSYS was applied to the mesh computational model.

- Grid independence

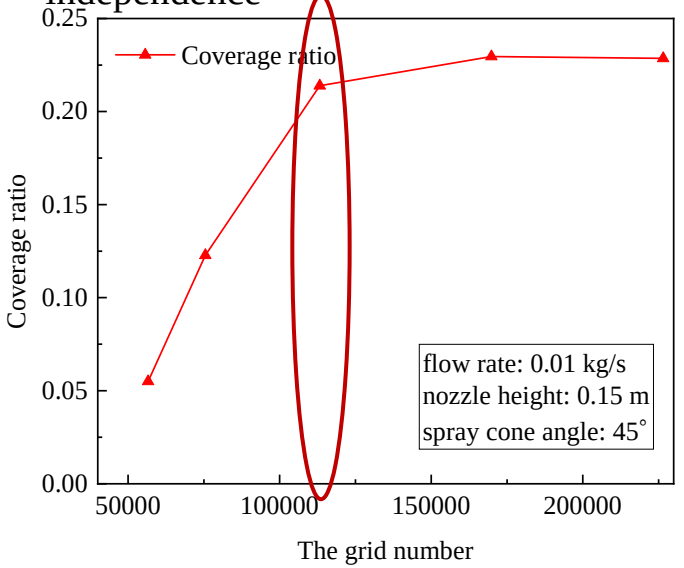


Fig.4.3.1 Coverage ratio at different numbers of grid

- The grid number of **113200** could be decided as the simulation number for further research.

中文要点｜网格与水膜分布验证

采用 **113,200** 个网格开展后续研究，并将模拟水膜分布与实验及 ImageJ 图像处理结果对比。

4. Development of CFD model for IEC considering water film formation with spray parameter analysis

4.4 Environment validation

- Conditions:
  - Q=0.01 kg/s;
  - Spray cone angle = 45°;
  - Nozzle height =150 mm

- Requirements:
  - Outermost transparent casing;
  - Sliding nozzles mounted on top

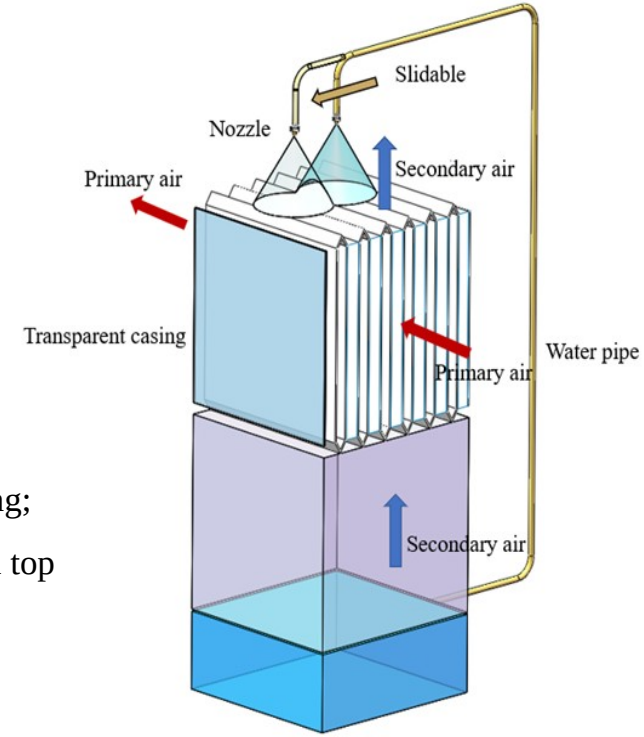


Fig.4.4.1 Experiment bench

- Validation result:

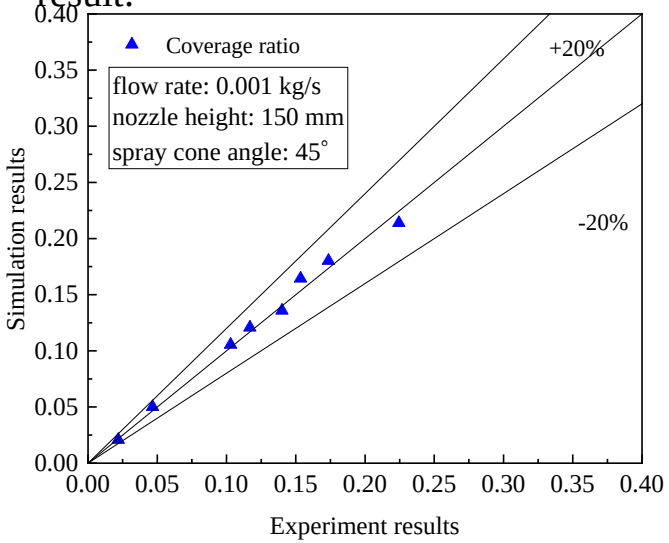


Fig.4.4.2 Validation results

- The maximum discrepancy is **7.1%**, which was acceptable for the further analysis.

中文要点｜水膜模型实验验证

在流量 **0.01 kg/s**、喷雾锥角 **45°**、喷嘴高度 **150 mm** 的验证工况下，最大偏差为 **7.1%**。

4. Development of CFD model for IEC considering water film formation with spray parameter analysis

4.4 Environment validation

- Experiment bench

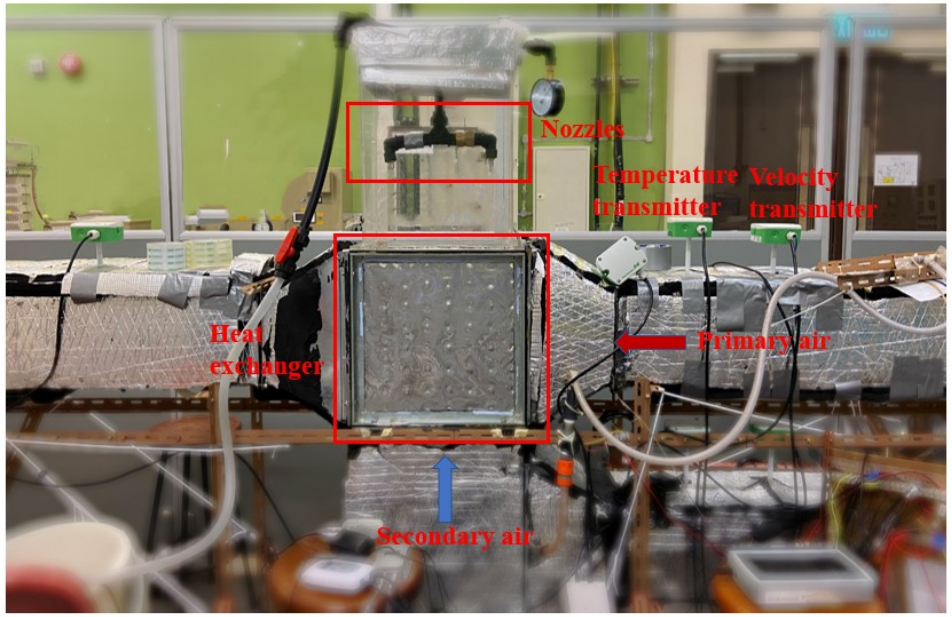


Fig.4.4.3 Validation results

- Validation result:

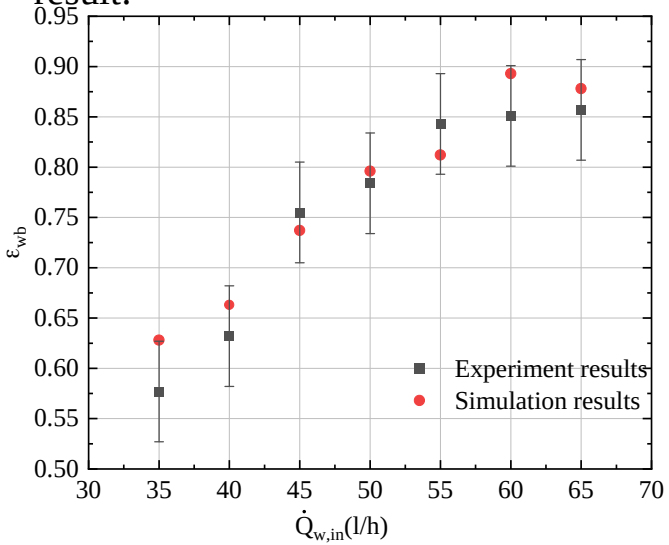


Fig.4.4.4 Validation results

- The maximum discrepancy is **8.2%**, which was acceptable for the further analysis.

中文要点｜系统性能实验验证

实验与模拟结果的最大偏差为 **8.2%**， 满足后续参数分析与系统性能研究的要求。

Appendix

- Spray characteristics

Theoretical mass flow rate is given by Lefebvre and McDonell

$$\dot{m}_{th} = \rho A_O \left(\frac{2\Delta P}{\rho}\right)^{0.5}$$

The average droplet diameter is obtained from the following equation:

$$d = 9.5a_f / (\Delta P_1 \sin(\alpha/2))$$

The correlation for spray cone angle based on inviscid theory (Abramovich, 1944) overpredicts the spray cone angle by 25%.

$$2\alpha = 3 \ldots \left(\frac{D_S}{\ldots}\right)^{-0.165} \ldots^{0.043} \ldots^{-0.065}$$

- The parameters are interrelated and cannot be adjusted individually.
- Maximise the film coverage and to balance the effects of each factor.

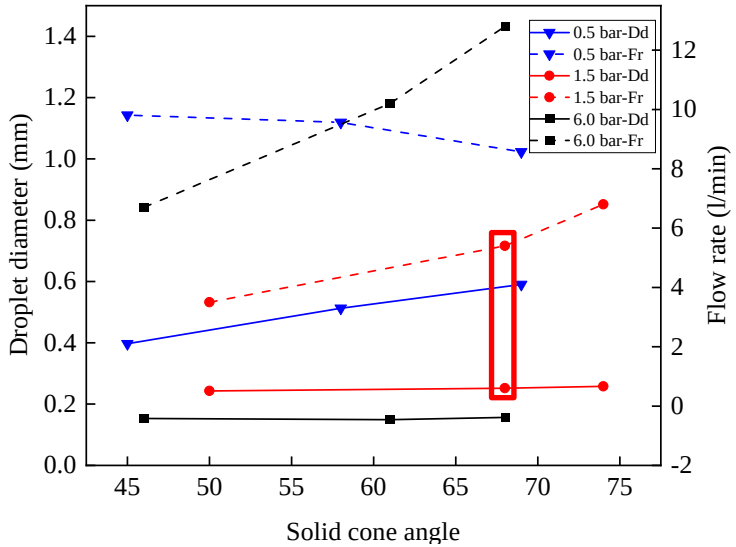


Fig. 8.4 Nozzle spray characteristics

- Ideal object: nozzle with a orifice diameter of 3.0 mm
- Operating condition: at a pressure of 1.5 bar, the spray

中文要点｜喷雾参数的关联性

流量、液滴直径和喷雾锥角相互关联，不能独立调节；优化时应综合平衡各因素对水膜覆盖的影响。

4. Development of CFD model for IEC considering water film formation with spray parameter analysis

4.5 Results and discussions

4.5.1 Influence for single nozzle

- Spray droplet diameter

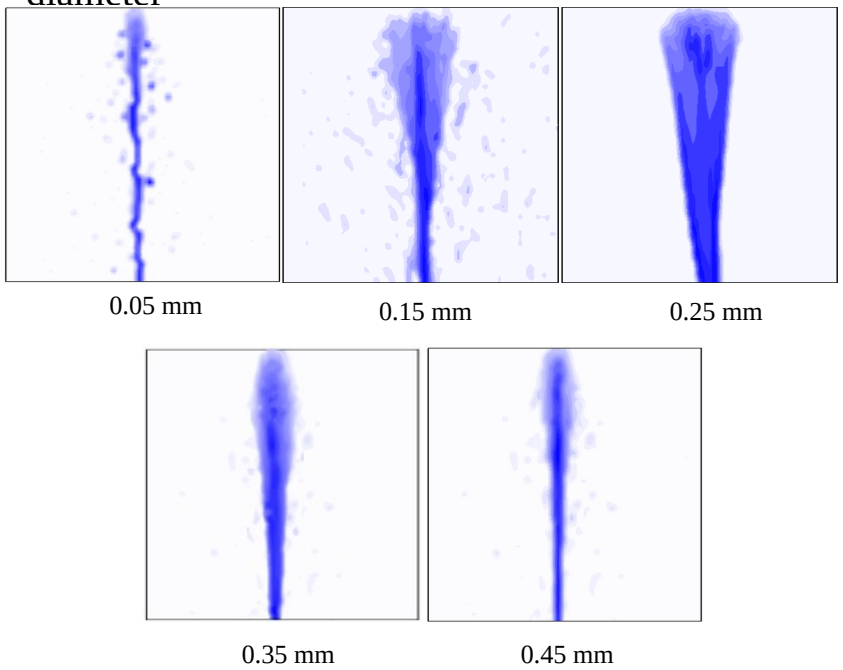


Fig. 4.5.1 Coverage area with different diameter

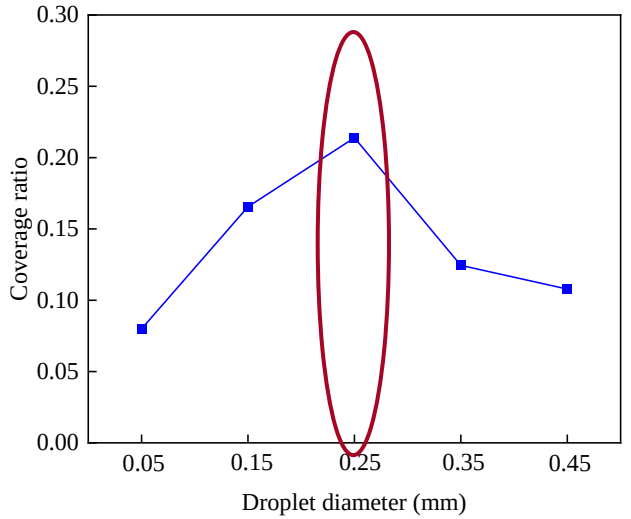


Fig. 4.5.2 Coverage ratio with different diameter

Droplet flow is **counter-current** to the airflow.

- Too small, be **blown away** before contact with the plate surface.
- Two large, **slide out of the wet channel** before making contact with the plate surface.

中文要点｜液滴直径的影响

液滴过小易被逆向气流吹离，过大则可能在接触板面前滑出通道；需选择适宜的液滴尺度。

4. Development of CFD model for IEC considering water film formation with spray parameter analysis

4.5.2 Influence for multiple nozzles

- Spray distance

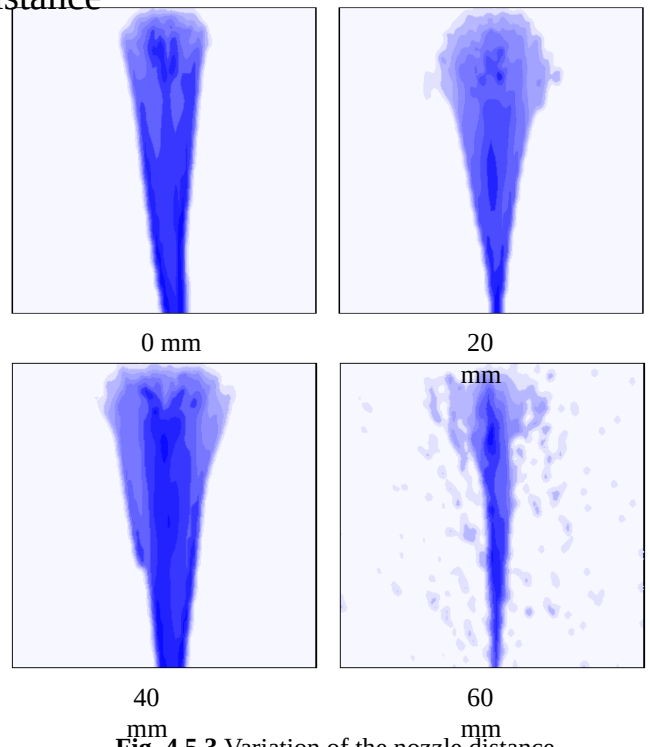


Fig. 4.5.3 Variation of the nozzle distance

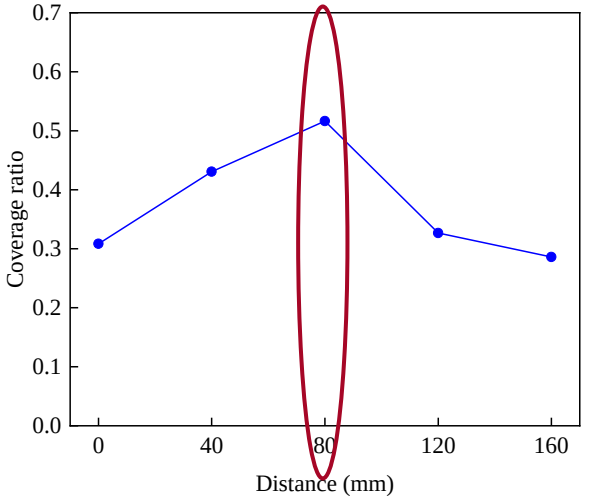


Fig. 4.5.4 Impact of nozzle gap on coverage factor

The best wetting situation of the plate surface occurs at a nozzle of **40 mm (1)** and the gap was **80 mm (2)**.

- The spray of nozzles are **too far** apart causing some droplets to **escape** from the boundary.
- While being **too close** caused a large number of spray droplets to **collide and splash** against each other.

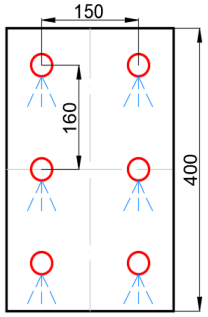
中文要点｜喷嘴间距的影响

间距过大易使液滴逸出边界，过小则加剧碰撞飞溅；应通过布置优化提高板面覆盖率。

4. Development of CFD model for IEC considering water film formation with spray parameter analysis

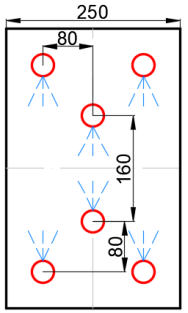
4.5.3 Influence on IEC performance

- Conventional scheme



Linear distribution

- Optimized scheme



Triangular distribution

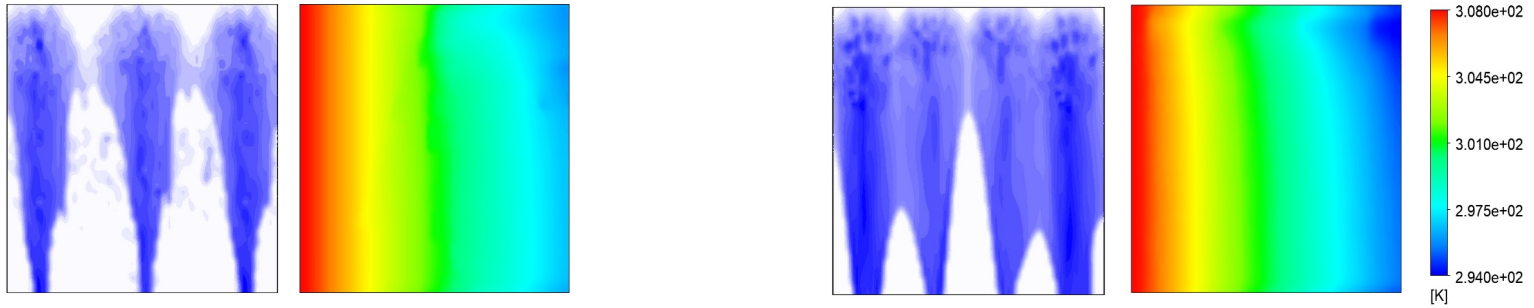


Fig. 4.5.5 Comparison between conventional and optimized schemes

- The water film coverage ratio of conventional and optimized schemes are **0.72 and 0.91**.
- **Most rapid temperature drop** at the **top** of the heat exchanger plate, due to the **widest area** covered by the water film and the **longest contact** between airflow and water flow.

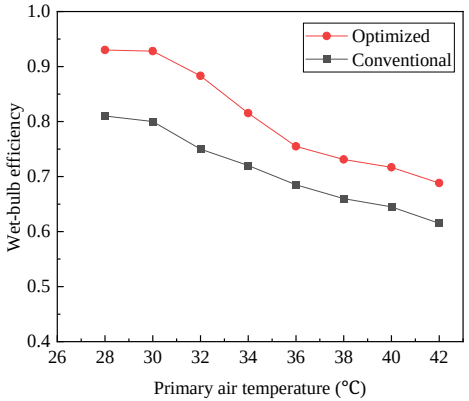
中文要点｜常规与优化方案对比

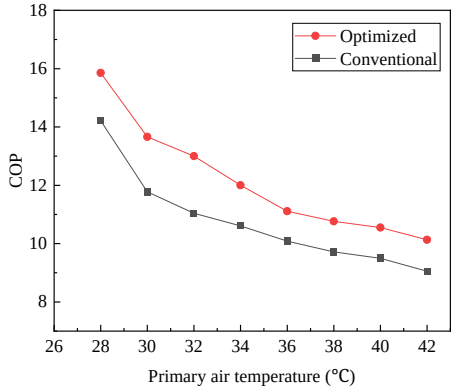
优化后水膜覆盖率由 **0.72** 提高至 **0.91**；板面顶部因水膜覆盖充分、气水接触时间长，降温更快。

4. Development of CFD model for IEC considering water film formation with spray parameter analysis

4.5.4 The effect of inlet parameter of primary air and secondary air

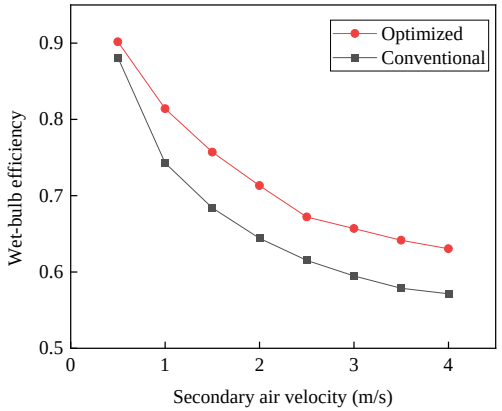
- Inlet temperature of primary air

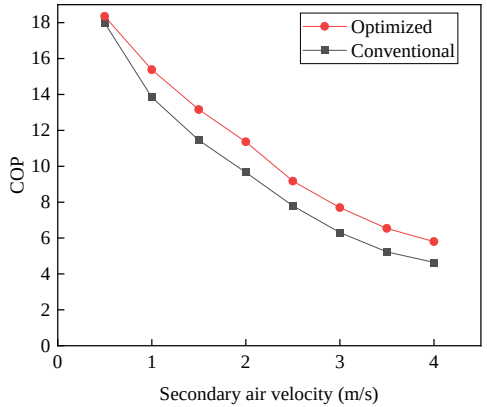




- An inlet air temperature of 32°C, with increases of 15.1% and 17.6% for ε_w and COP.

- Inlet velocity of secondary air





- An inlet air velocity of 2 m/s, with increases of 10.4% and 14.9% for $_w$ and COP.

中文要点｜入口空气参数的影响

在 **32 ℃** 工况下，湿球效率与 COP 分别提高 **15.1%** 和 **17.6%**； **2 m/s** 工况下分别提高 **10.4%** 和 **14.9%** 。

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6. Performance analysis and nozzle setting optimization of IEC for energy recovery in data centers
7. Conclusions and recommendations for future work

中文要点｜第五部分： ANN–GA 优化

基于实验数据建立人工神经网络（ANN），结合遗传算法（GA）开展喷淋策略的多目标优化。

5. Performance assessment and optimization of water spray strategy for IEC based on ANN modeling and GA

5.1 Concept and model set up

- Data preparation
  - The **six parameters** of the spray system that affect the performance of IEC can be identified in the table.
  - Evaluation indexes: **wet-bulb efficiency** and **pressure drop**.
- Collection of experimental data

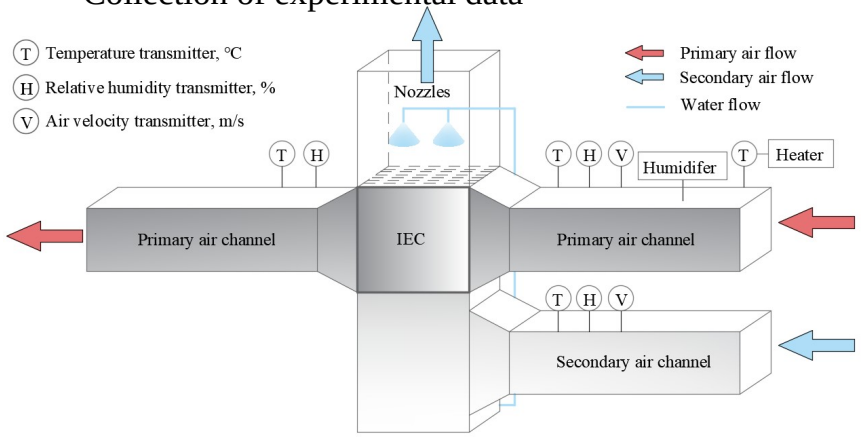


Fig.5.1.1 Schematic diagram of the IEC system

Table 5.1.1 Solid nozzle selection

No.	Input parameters	Minimum	Maximum
1	Nozzle pressure	0.05 MPa	0.2 MPa
2	Nozzle spacing	0 mm	300 mm
3	Nozzle aperture	2.5 mm	6 mm
4	Nozzle number	0	15
5	Spraying density	0.1 kg/(s·m^2)	2 kg/(s·m^2)
6	Air-water ratio (weight ratio)	1	8

- Conducted on a laboratory-scale prototype of the IEC system
- Components: cross-flow heat exchanger, nozzles and water pump.
- Keep the other basic parameters of the IEC constant, change the six main parameters, and record the effects.

中文要点｜实验数据与评价指标

改变六项主要喷淋参数，采集实验数据；以湿球效率与压降作为模型预测和优化的评价指标。

5. Performance assessment and optimization of water spray strategy for IEC based on ANN modeling and GA

5.2 Design of model setting

- A **3-layer** network with a hidden layer comprising **14 neurons**, can effectively capture the relationship between the input and output variables.
- The dataset could be divided into three sections: the training dataset, validation dataset, and testing dataset, with portions of **70%**, **15%**, and **15%**, correspondingly

➢ Equations describe the model

- Input layer

$$X = (x_1, x_2, \dots, x_i, x_n)^T$$

- Hidden layer

$$net_j = w_{1j} \cdot x_1 + \cdots + w_{nj} \cdot x_n$$

$$h_j = f\left(net_j - a_j\right)$$

$$H = \left(h_1, h_2, \dots, h_j, \dots, h_i\right)^T$$

- Hidden layer

$$net_k = w_{1k} \cdot h_1 + \cdots + w_{lk} \cdot h_l$$

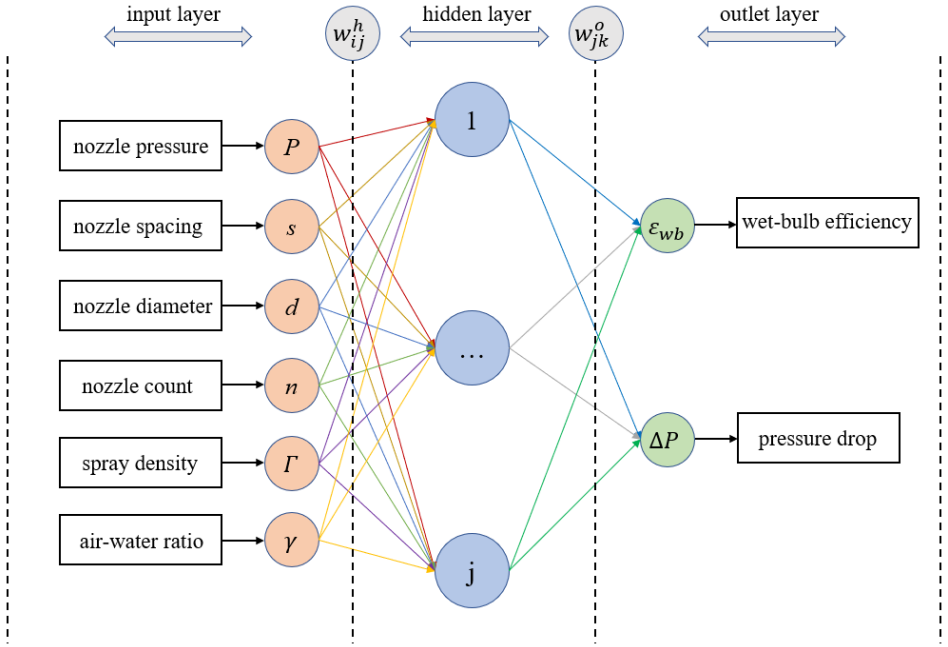


Fig.5.2.1 Structure of the IEC-ANN model

中文要点｜神经网络结构

采用三层网络，隐含层含 **14** 个神经元；训练集、验证集与测试集的比例分别为 **70%** 、 **15%** 和 **15%** 。

5. Performance assessment and optimization of water spray strategy for IEC based on ANN modeling and GA

5.3 Model validations

- Evaluation metrics of neural network

- Coefficient of determination (r^2)

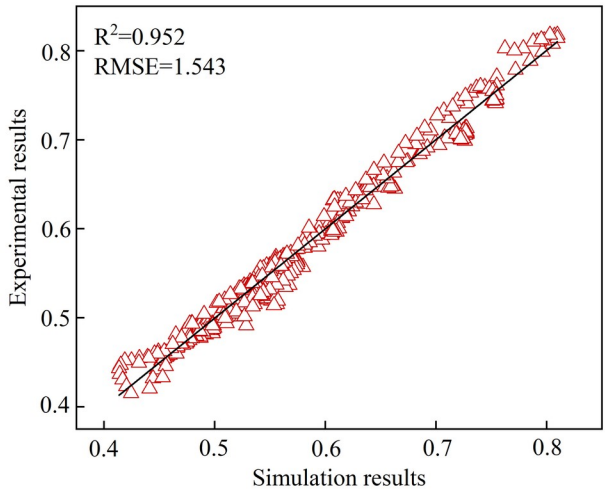
$$r^2 = \frac{[n\sum_{k=1}^{n} Y_P(k)\,Y_T(k) - \sum_{k=1}^{n} Y_P(k)\sum_{k=1}^{n} Y_T(k)]^2}{[n\sum_{k=1}^{n} Y_P(k)^2 - [\sum_{k=1}^{n} Y_P(k)]^2][n\sum_{k=1}^{n} Y_P(k)^2 - [\sum_{k=1}^{n} Y_T(k)]^2]}$$

- Root mean square error (RMSE)

$$RMSE = \sqrt{\frac{\frac{n}{i=k}\left(Y_P(k) - Y_T(k)\right)^2}{n}}$$

- Validation results

- Validation



- V

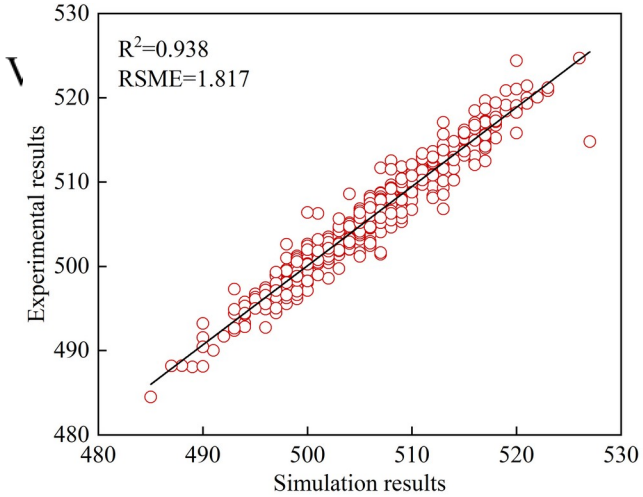


Fig.5.3.1 Validation results of the proposed IEC-ANN model

中文要点｜ ANN 模型验证

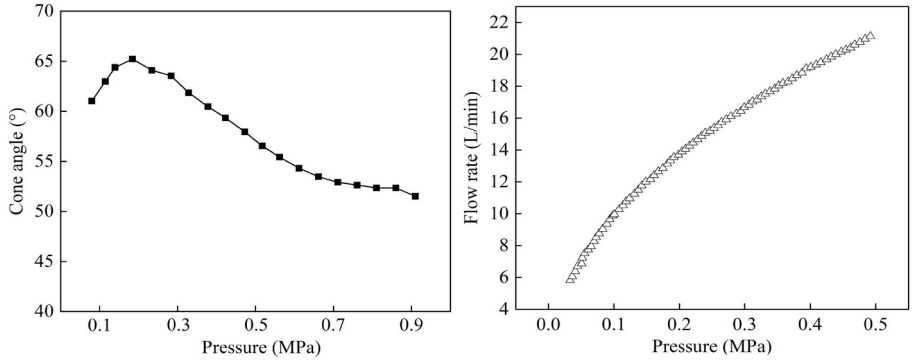
通过神经网络评价指标及预测值与实验值的对比，检验所建立 IEC–ANN 模型的预测能力。

5. Performance assessment and optimization of water spray strategy for IEC based on ANN modeling and GA

5.4 Results and discussions

5.4.1 Effect of variations in a single factor

- Nozzle characteristics parameters



(a) Effect of pressure on cone angle (b) Effect of pressure on flow rate

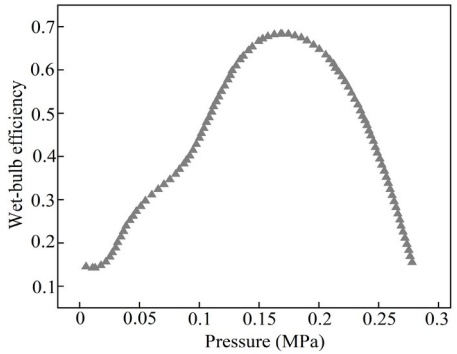
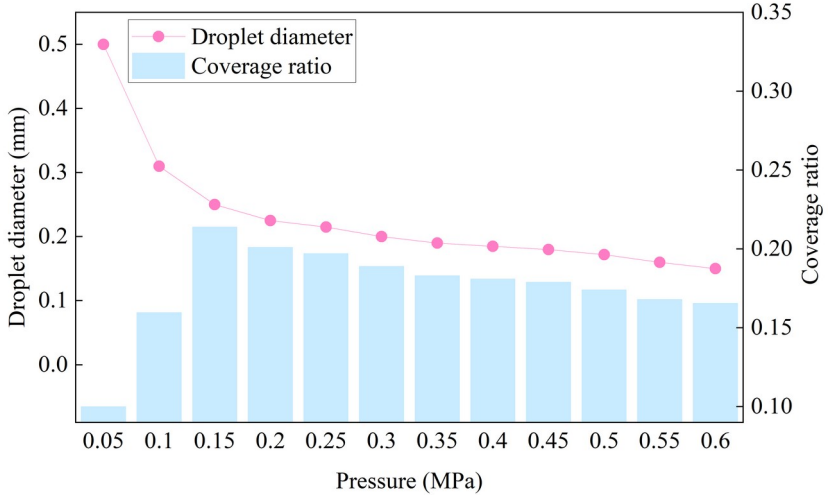


Fig.5.4.1 The effect of nozzle pressure variations

- Effect of pressure on droplet diameter and coverage ratio



(a) The peak value of 67° is achieved at a pressure of **0.15 MPa**.

(b) The increasing trend of spray flow rate with the rise in pressure.

中文要点｜喷嘴压力的影响

压力改变会同时影响喷雾锥角、流量和液滴直径，进而影响水膜覆盖和系统性能。

5. Performance assessment and optimization of water spray strategy for IEC based on ANN modeling and GA

5.4 Results and discussions

5.4.1 Effect of variations in a single factor

- Nozzle spacing
- Nozzle aperture
- Nozzle count

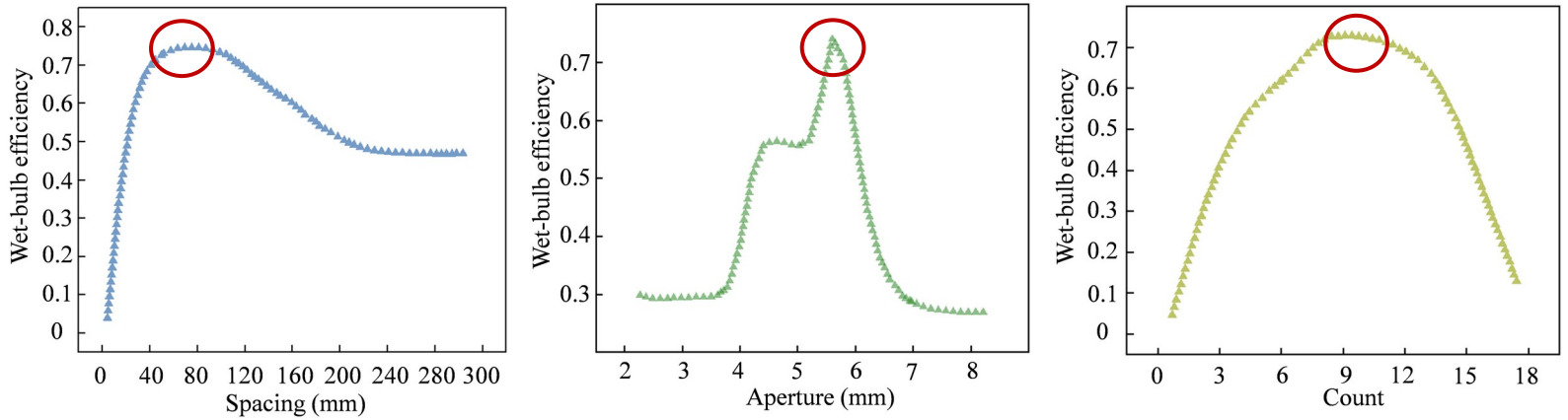


Fig.5.4.2 The effect of three other nozzle parameters variations

- The peak value of wet-bulb efficiency is achieved at a spacing of **80 mm**.

中文要点｜喷嘴几何参数的影响

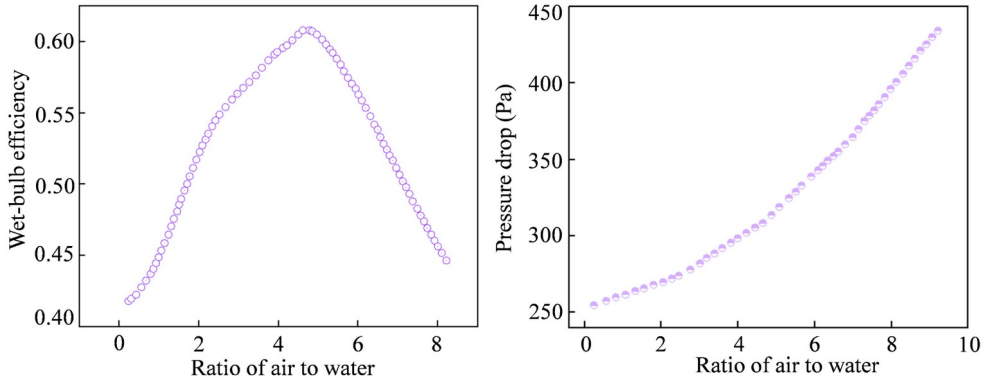
喷嘴间距、孔径和数量共同影响喷淋分布，应结合润湿效果与运行阻力进行选择。

5. Performance assessment and optimization of water spray strategy for IEC based on ANN modeling and GA

5.4 Results and discussions

5.4.1 Effect of variations in a single factor

➢ Operation characteristics parameters



➢ Ratio of air to water

- Initial stage: improves the convective heat and mass transfer coefficient, thus optimising the ε_{wb}.
- Later stage: disrupts the formation of a water film, increasing the pressure drop

➢ Spray density

- Lower stage: Uneven water film distribution, presence of dry areas
- Higher stage: thicker water film lead to clogging of secondary air channel, increasing air resistance.

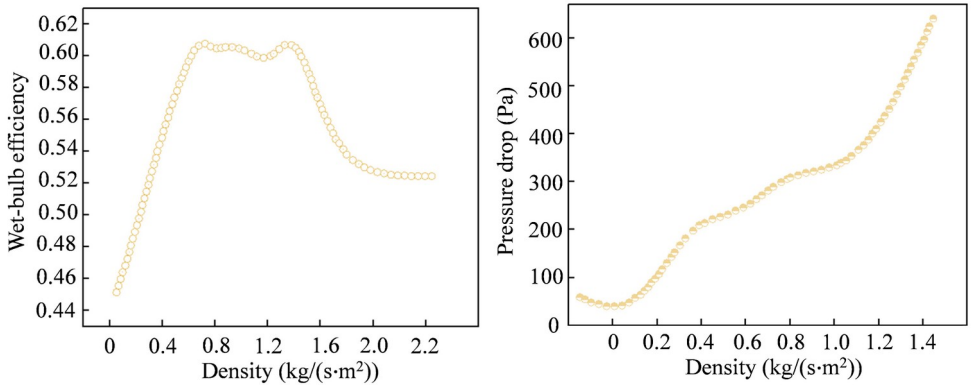


Fig.5.4.4 The effect of operation parameters on pressure drop

中文要点｜喷淋密度的影响

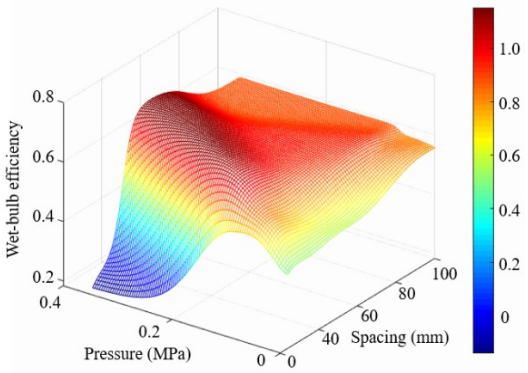
喷淋密度低时水膜不均、存在干区；过高时水膜增厚，可能堵塞二次空气通道并增大空气阻力。

5. Performance assessment and optimization of water spray strategy for IEC based on ANN modeling and GA

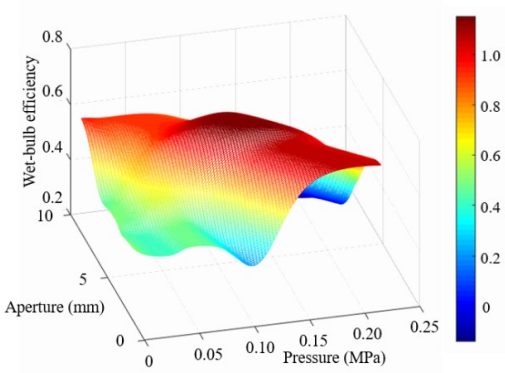
5.4 Results and discussions

5.4.2 Combined effect of variations in two factors

➢ Spacing and

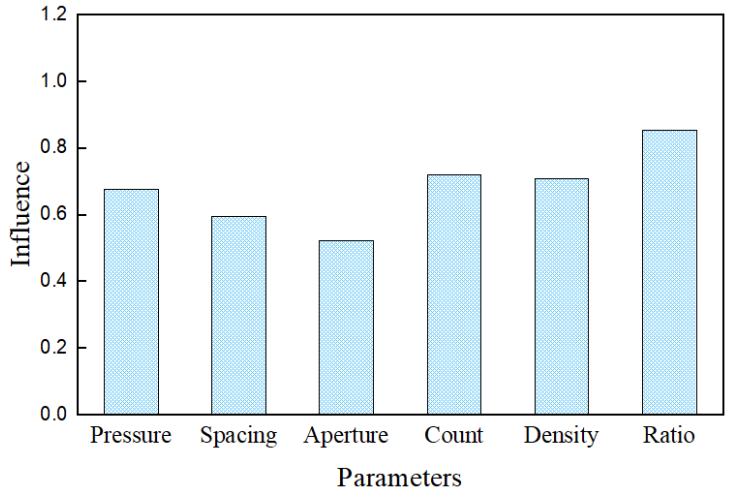


➢ Aperture and pressure



- Small spacing: increasing the pressure rapidly brings the IEC performance to its peak and then decreases.
- Low pressure: aperture ε_{wb}

➢ The importance of different design



- Ratio < Count < Density < Pressure < Spacing < Aperture
- The smallest impact factor with a correlation

中文要点｜双因素耦合与参数重要性

分析间距—压力、孔径—压力的耦合影响，并比较各设计参数对系统性能的重要性。

5. Performance assessment and optimization of water spray strategy for IEC based on ANN modeling and GA

5.4 Results and discussions

5.4.3 Optimization of IEC-ANN model based on Genetic algorithm

- Genetic algorithm
  - Four key aspects: determining the basic operating parameters, chromosome encoding and decoding, evaluating individual fitness, and implementing the three genetic operators.

- Multi-objectives optimization
  - Optimization targets: wet-bulb efficiency ↑ and pressure drop ↓

    Minimizing power consumption & **Maximizing** cooling performance.
  - The optimization function comprises of both a fitness as well as a constraint function.

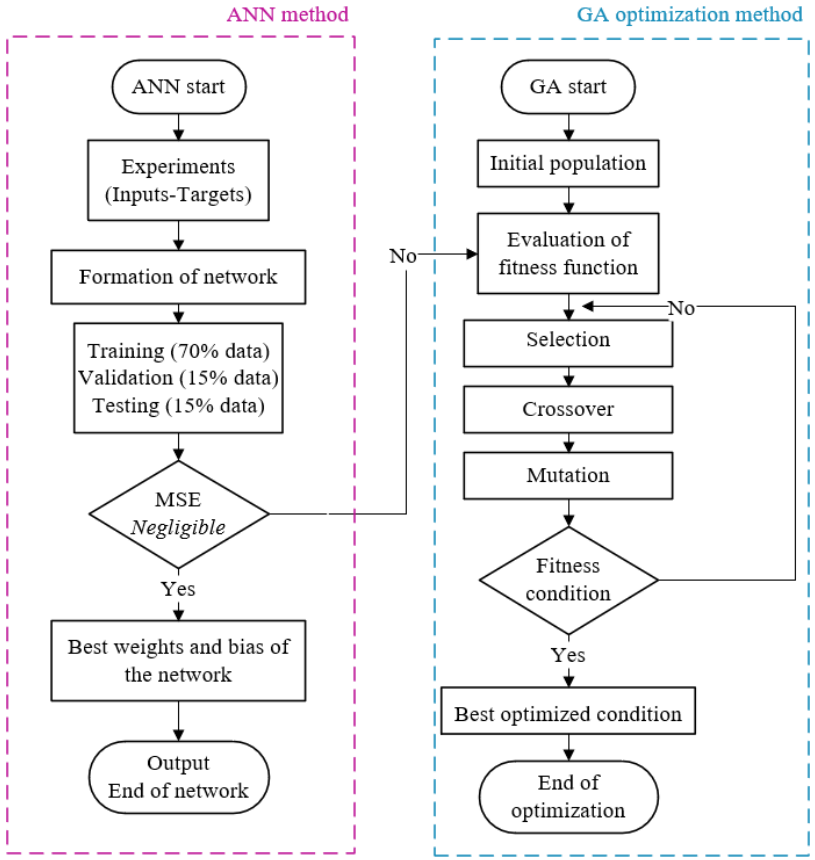


Fig.5.4.3 The flow of GA-ANN optimization

$$J(P, s, d, n, \Gamma, v) = W_1 \frac{\varepsilon_{wb}}{P_{wb}} + W_2 \frac{\Delta P}{P_{dB}}$$

中文要点｜遗传算法多目标优化

以提高湿球效率、降低压降为目标，通过适应度评价及遗传操作，平衡制冷性能与运行能耗。

5. Performance assessment and optimization of water spray strategy for IEC based on ANN modeling and GA

5.4 Results and discussions

5.4.3 Optimization of IEC-ANN model based on Genetic algorithm

- Results derived from GA

Design weight			Design variables						Objectives		
No.	W_1	W_2	Pressure (MPa)	Spacing (mm)	Aperture (mm)	Count	Density (kg/(s·m^2))	Velocity (m/s)	ε_{wb}	ΔP	COP
1	0.75	0.25	0.15	60 mm	6 mm	14	1.5	3.5	0.86	328	10.35
2	0.5	0.5	0.15	85 mm	6 mm	11	0.83	2.8	0.85	210	12.4
3	0.25	0.75	0.15	95 mm	6 mm	8	0.31	1.7	0.77	97	7.71
Base system			0.1	80 mm	6 mm	8	0.44	2.0	0.81	117	9.52

- An **equal-weight distribution** yields the best conditions for optimization of IEC performance.
- Under this condition, the optimal operating parameters are fixed, which is an increase of **23.2%** in comparison with the base system.

中文要点｜ ANN–GA 优化结果

等权重分配获得最佳综合优化效果；确定最优运行参数后，相比基准系统，性能提升 **23.2%** 。

Outline

1. Research background and literature review
2. Research gaps and objectives
3. Development of FDM model for spray distribution optimization of IEC
4. Development of CFD model for IEC considering water film formation with spray parameter analysis
5. Performance assessment and optimization of water spray strategy for IEC based on ANN modeling and GA
6. Performance analysis and nozzle setting optimization of IEC for energy recovery in data centers
7. Conclusions and recommendations for future work

中文要点｜第六部分：数据中心应用

针对数据中心用菱形 IEC，研究喷嘴布置、板面强化措施及能量回收性能。

6. Performance analysis and nozzle setting optimization of IEC for energy recovery in data centers

6.1 Model description

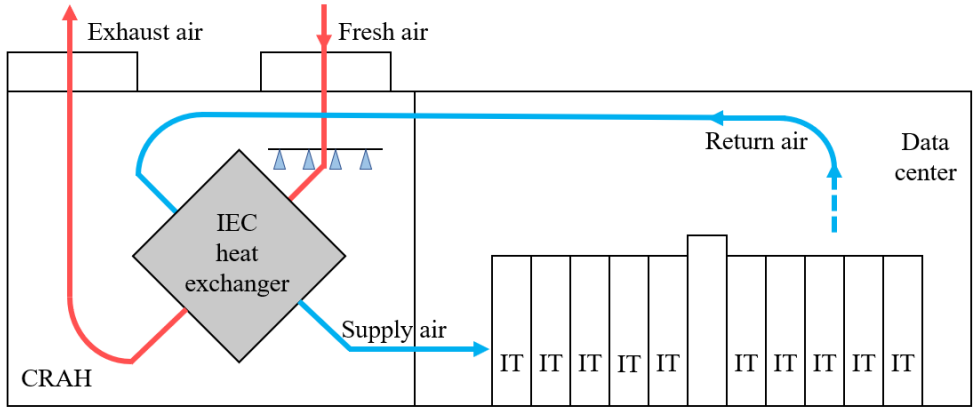


Fig.6.1.1 A typical IEC air-conditioning system of DC

- Internal circulation
- Fresh air as the working air
- Indoor return air as supply air
- Separated channels avoid the cross-contamination.

- Diamond shaped IEC: compactness and flexible arrangement
- Space claim in **ANSYS** work bench

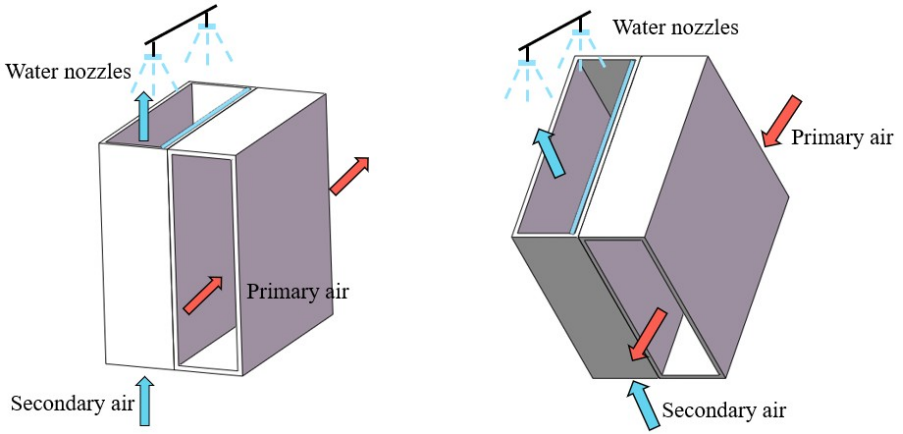


Fig. 6.1.2 Two types of IEC models set up in work bench

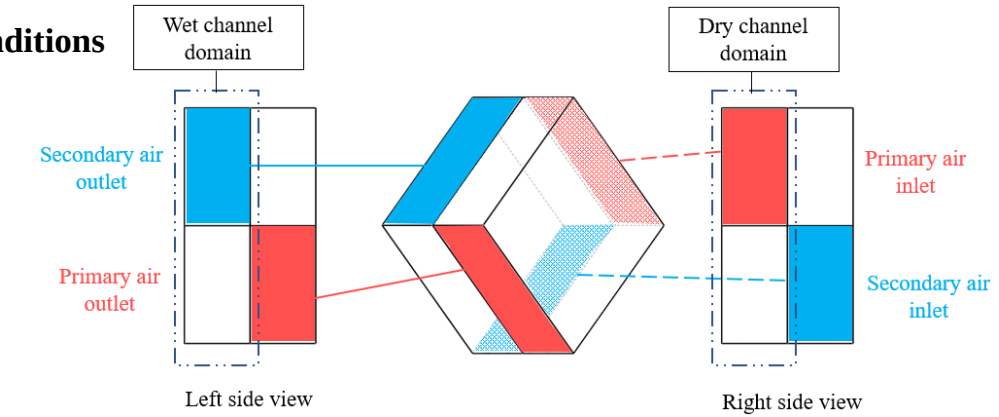
中文要点｜菱形 IEC 与内循环模式

新风作为工作空气，室内回风作为送风；独立通道避免交叉污染，菱形结构兼具紧凑性与布置灵活性。

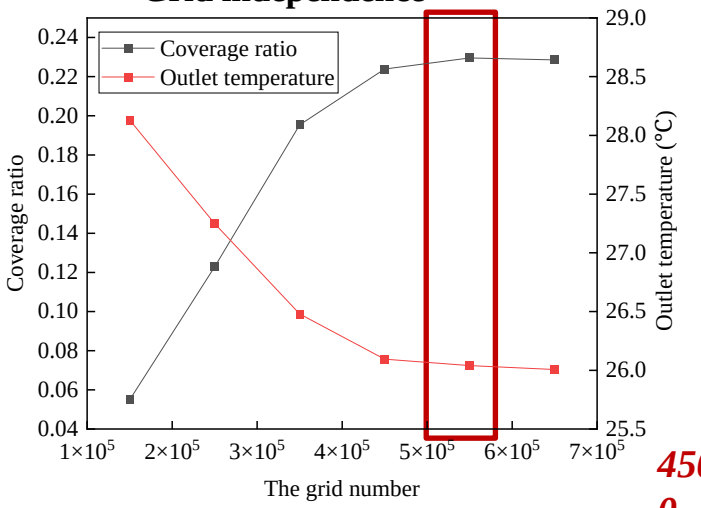
6. Performance analysis and nozzle setting optimization of IEC for energy recovery in data centers

6.2 Model details

➢ **Boundary conditions**

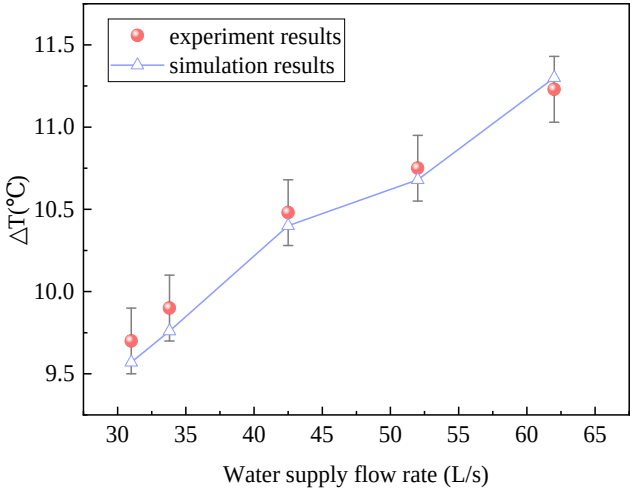


➢ **Grid independence**



450,000

➢ **Model validation**



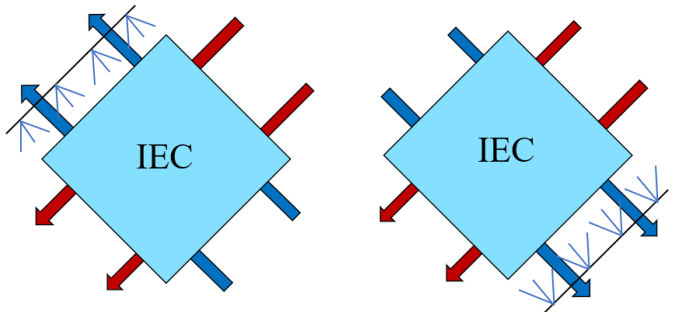
10 %

中文要点｜模型边界与验证

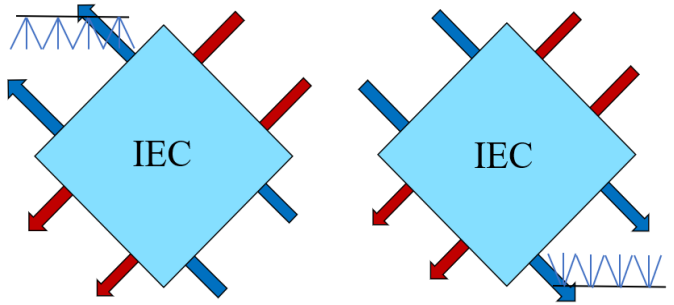
明确边界条件，并通过网格无关性和实验对比，检验菱形 IEC 模型的可靠性。

6. Performance analysis and nozzle setting optimization of IEC for energy recovery in data centers

6.3 Four schemes of nozzle configurations



(a) Top configuration (b) Bottom configuration



(c) Upper side configuration (d) Lower side configuration

➢ Determination of the test water flow rate

- The **wet-bulb efficiency** grows with the increase of **water supply flow rate**.
- and stabilizes at a maximum value when the flow rate arrives **65 L/s.**

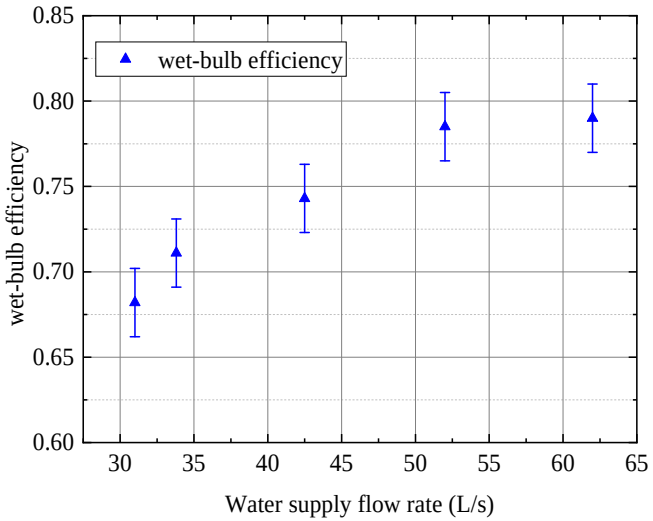


Fig. 6.3.1 The determination of test spray flow rate

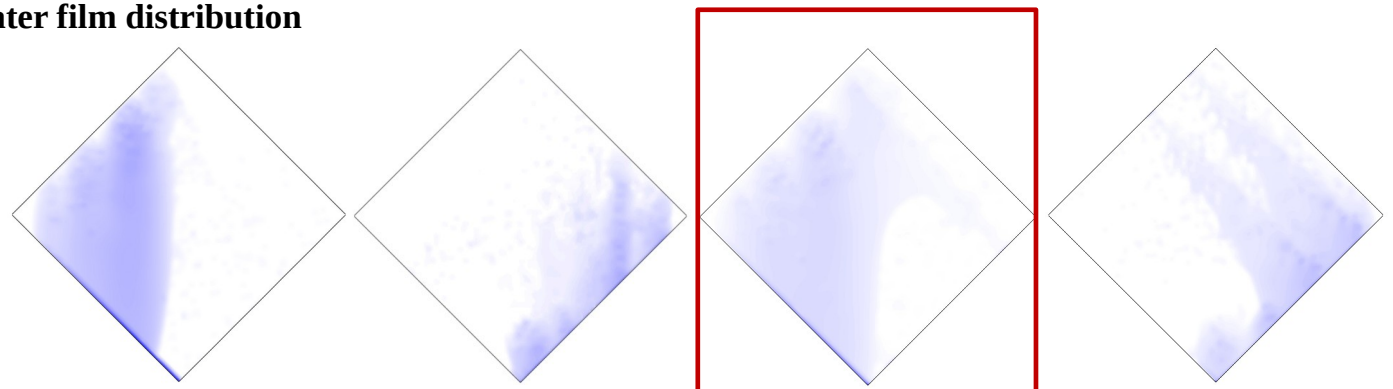
中文要点｜四种喷嘴布置方案

比较顶部、底部、上侧与下侧布置，并通过供水流量分析确定实验条件。流量数值及单位沿用原文。

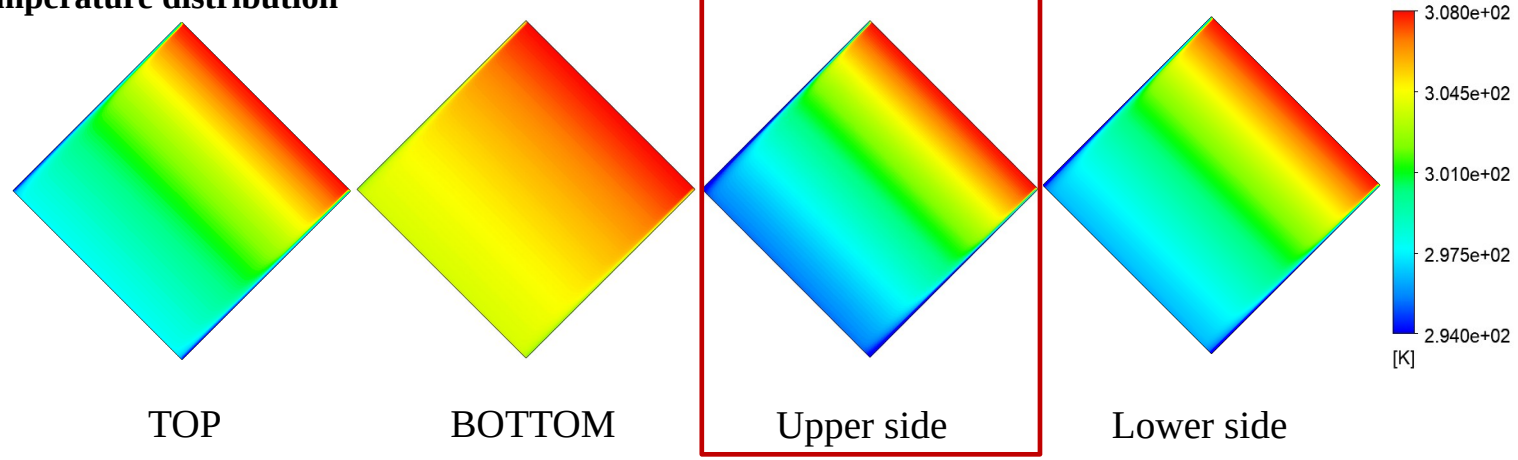
6. Performance analysis and nozzle setting optimization of IEC for energy recovery in data centers

6.3 Four schemes of nozzle configurations (aluminium)

- **Water film distribution**



- **Temperature distribution**



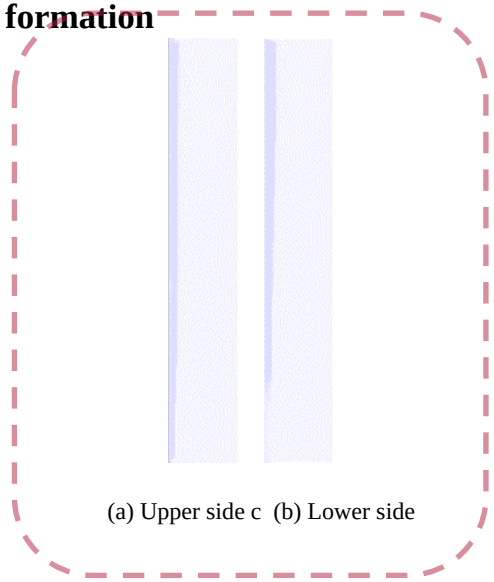
中文要点｜水膜与温度分布对比

对比铝板 IEC 在四种喷嘴布置下的水膜及温度分布，分析布水位置对换热过程的影响。

Appendix

Comparison of configurations of axial characteristics

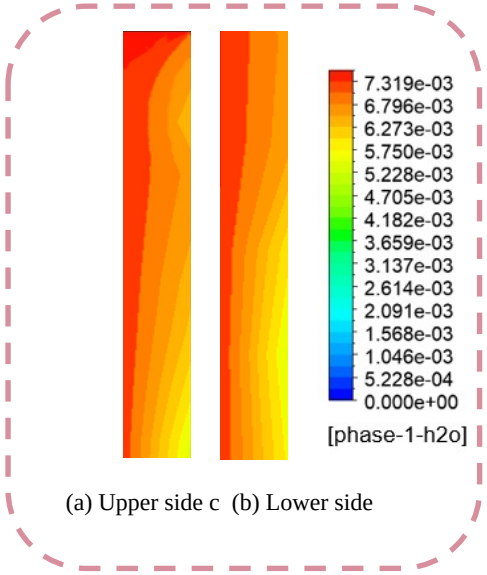
➢ **Film formation**



(a) Upper side c (b) Lower side

- More smooth and diffuse continuous water film.
- Incomplete water film coverage due to aggregation from **uneven gravity distribution**.

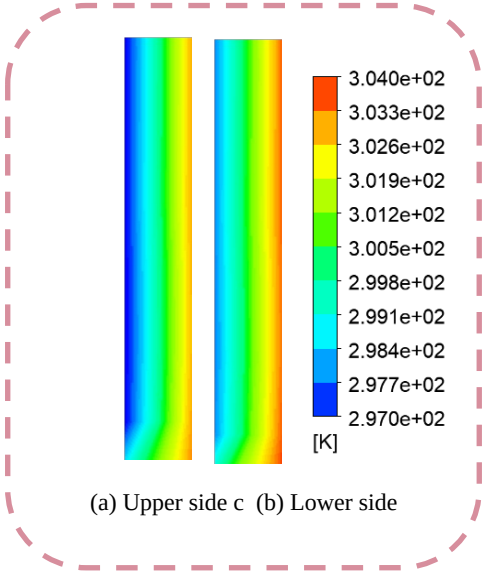
➢ **water vapor content distribution**



(a) Upper side c (b) Lower side

- The water vapor concentration significantly **higher**.
- The water membrane transfers more mass to the airflow.

➢ **Temperature distribution**



(a) Upper side c (b) Lower side

- More temperature drop.
- The temperature drop follows the same trend as the increase in water vapour concentration.

中文要点｜轴向分布特征

连续、均匀的水膜有利于传热传质；局部聚集与覆盖不足会限制水分向气流传递及降温效果。

Appendix

➢ **Specific gaps in the four configurations**

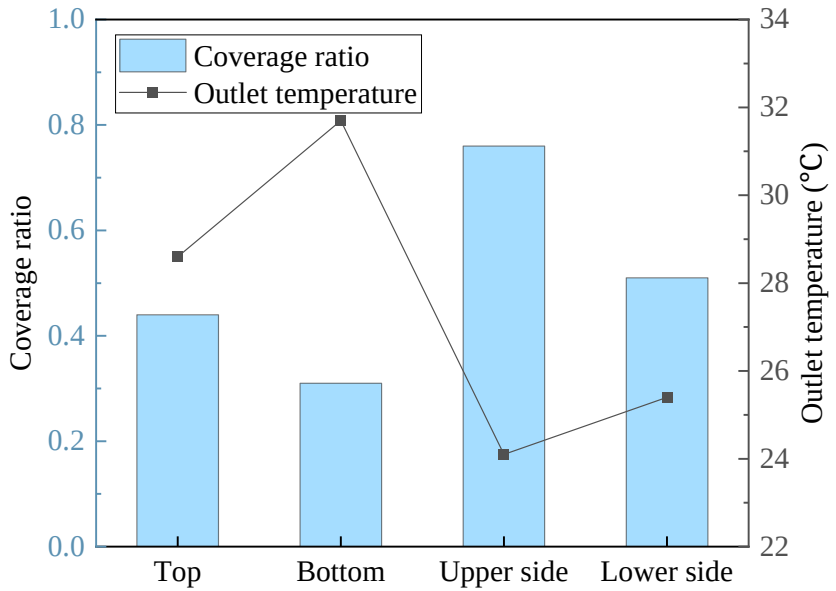


Fig. 8.5 The coverage ratio and outlet temperature of four configurations

- The **temperature drop** follows the same trend as **the water film coverage coefficient**.
- The best results are achieved in the **upper side** configuration, with a **59.2%** and **27.4%** improvement compared to the worst one.

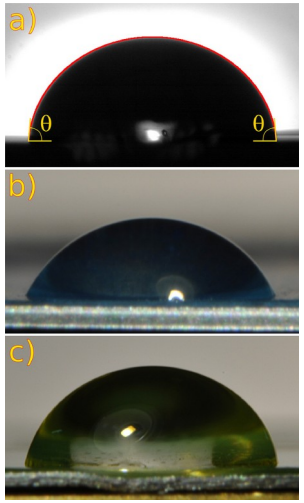
中文要点｜四种布置的性能差异

上侧布置效果最佳；原文两项指标相对最差方案分别提升 **59.2%** 和 **27.4%**，温降与水膜覆盖呈同向变化。

6. Performance analysis and nozzle setting optimization of IEC for energy recovery in data centers

6.4 Approaches for cooling performance enhancement

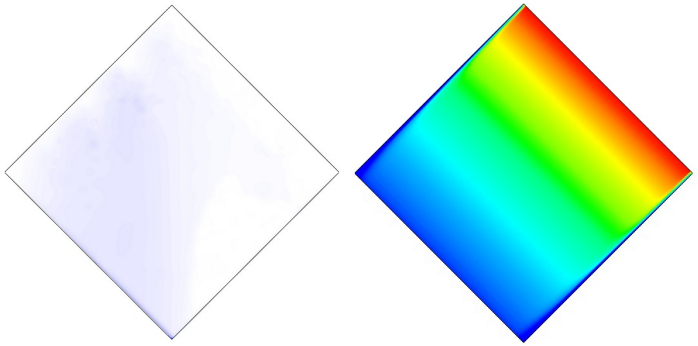
- **Hydrophilic coating (Epoxy coating-Hydrophilic lacquer)**



Contact angle

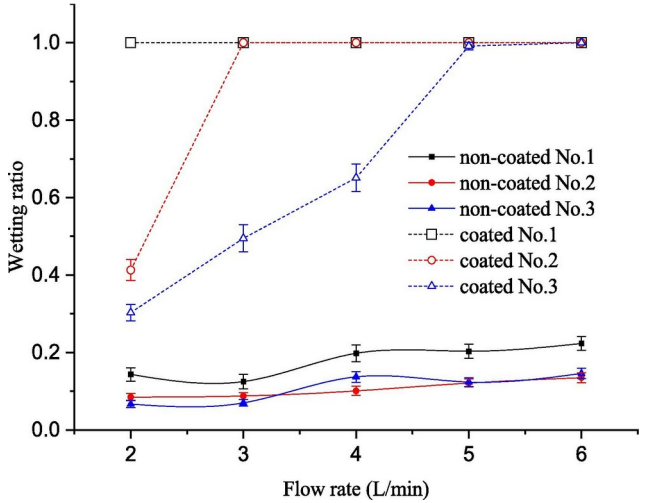
77.8°

57.8°



(a) Water film distribution (b) Temperature distribution

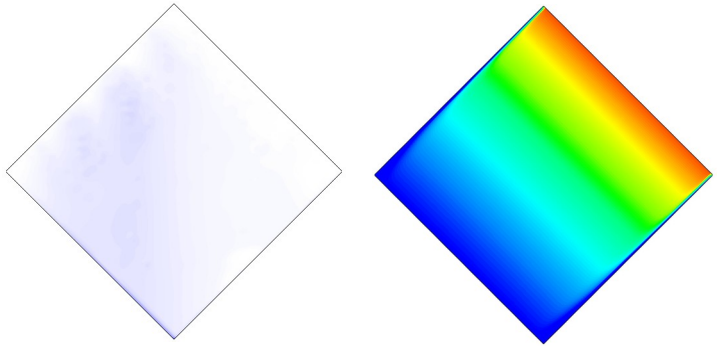
- **Fiber coating (70% rayon + 30% polyester textile)**



Contact angle

0°

diffusion ≈



(a) Water film distribution (b) Temperature distribution

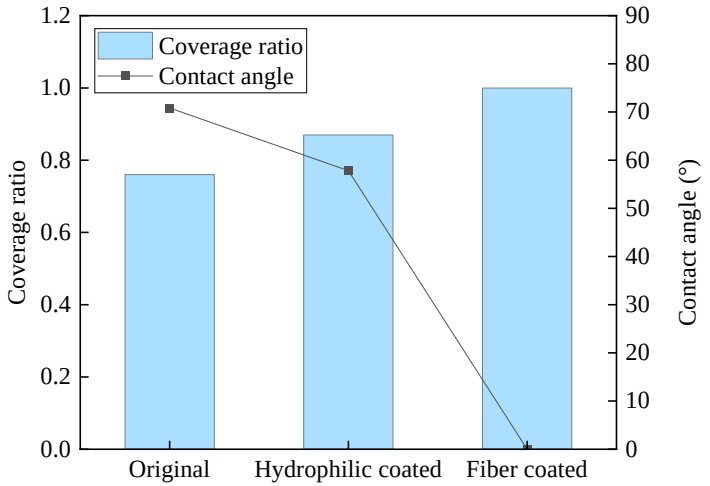
中文要点｜板面强化措施

比较亲水涂层与纤维涂层，通过降低接触角、促进水膜扩散，提高换热板的润湿性能。

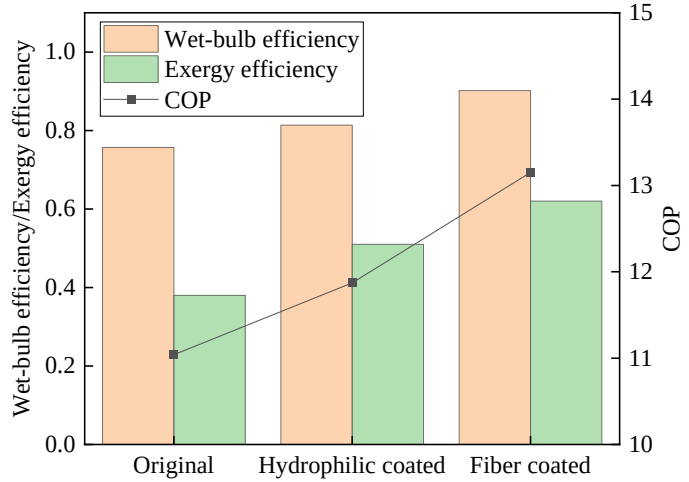
6. Performance analysis and nozzle setting optimization of IEC for energy recovery in data centers

6.5 Assessment of the degree of enhancement

➢ **Effect on coverage ratio and contact angle**



➢ **Comparison of the three performance indexes**



- Both the **hydrophilic** and **fiber** coating increase the wetting ratio by decreasing the contact angle
- Besides, the **fiber** coating also increases the diffusion rate of the water film compared to the hydrophilic one, thus achieving a **wetting ratio** close to **1**, which is regarded as **complete wetting**.
- The **wet-bulb efficiency** of the IEC was optimized for both the increased **hydrophilic** and **fiber** coating on the plate surface, by **5.7%** and **14.5%** respectively.
- The energy saving by pump resulting in a further increase for fiber coating in the **COP** value of **16.1%.**

中文要点｜强化效果评价

亲水与纤维涂层分别使湿球效率提高 **5.7%** 和 **14.5%**；纤维涂层还可节省泵耗，使 COP 提高 **16.1%**。

Outline

1. Research background and literature review
2. Research gaps and objectives
3. Development of FDM model for spray distribution optimization of IEC
4. Development of CFD model for IEC considering water film formation with spray parameter analysis
5. Performance assessment and optimization of water spray strategy for IEC based on ANN modeling and GA
6. Performance analysis and nozzle setting optimization of IEC for energy recovery in data centers
7. Conclusions and recommendations for future work

中文要点｜研究结论与未来展望

总结 FDM、CFD 与智能优化研究结果，并讨论模型适用性、运行控制及跨气候条件分析的拓展方向。

7. Conclusions and recommendations for future work

7.1 Conclusions

1. We developed a FDM model to simulate the spray distribution above heat exchanger.

•For the IEC with 400 mm length and width, the optimal nozzle arrangement scheme is the scheme with four nozzles separated by 160 mm along the centerline.

•With the same water flow rate supplied (220 L/h), the optimized centerline scheme could increase the wettability factor of the IEC plate surface from 0.48 to 0.89, and the COP is increased up to 16%.

2. We proposed a novel 3D CFD model to investigate the water film inside wet channels, and the cooling performance of the whole IEC system.

•Optimal nozzle parameters for maximum coverage were determined to be a droplet diameter of 0.25 mm, a pressure of 1.5 bar, a flow rate of 5.4 L/min, and a spray cone angle of 68°.

•The refined IEC water distribution system employs a triangular nozzle arrangement based on distance analysis. This optimized configuration enhances water film coverage by 20.9% and significantly improves wet-bulb efficiency and COP by 15.1% and 17.6%, respectively.

中文要点｜主要研究结论

FDM 优化使润湿系数由 **0.48** 升至 **0.89**， COP 提高 **16%**； CFD 优化方案使湿球效率与 COP 提高 **15.1%** 和 **17.6%**。

7. Conclusions and recommendations for future work

7.1 Conclusions

3. We trained a ML-based prediction model to estimate the performance of the IEC system, and GA was applied to optimize the IEC-ANN model.

- The optimal parameters were found to be: nozzle pressure of 0.15 MPa, spacing of 85 mm, aperture of 6 mm, count of 11, spray density of 0.85 kg/(s·m2), and air velocity of 2 m/s.
- The wet-bulb efficiency is 0.85, the pressure drop is 210 Pa, and the COP is 12.4, which represents a 23.2% improvement over the baseline system.

4. We developed a CFD model based on the diamond-shaped IEC to explore the air-water flow, heat transfer, and mass transfer capabilities of the IEC as a heat recovery unit in data centers.

- The nozzles are installed as upper side configuration when the water film distribution area in the wet channel is the largest and the temperature drop is the best, with 59.2% and 27.4% improvement over the worst case.
- Benefit from the addition of hydrophilic coatings and fibre coatings, the water film coverage and COP are

中文要点｜研究结论（续）

ANN–GA 优化方案的性能提升为 **23.2%**；菱形 IEC 中，纤维涂层可改善润湿性并使 COP 提高 **16.1%**。

7. Conclusions and recommendations for future work

7.2 Future work

1. The primary focus of this study is on the common cross-flow plate-type IEC, which may not be applicable to other specialized types, such as tubular IEC. Therefore, it is necessary to conduct extensive simulations and experimental research on **various types of IEC** in the future to generalize the findings.

2. For the **three operating modes** of the IEC system in data centers, it is crucial to regulate the start and stop cycles of the water spray system to meet indoor requirements while minimizing the energy consumption. Therefore, it is necessary to further control the activation states of the three modes based on climatic conditions.

3. Future work will involve the development of ANN models for different spray parameters under **various climatic conditions** to achieve a more comprehensive analysis and optimization of the IEC system.

中文要点｜未来研究方向

拓展至不同结构的 IEC；根据气候条件优化数据中心运行模式与喷淋启停；建立跨气候条件的 ANN 预测模型。

Academic contributions

Published papers:

[1] X. Ma, W. Shi, and H. Yang, “Study on water spraying distribution to improve the energy recovery performance of indirect evaporative coolers with nozzle arrangement optimization,” Applied Energy, vol. 318, 2022.

[2] X. Ma, W. Shi, and H. Yang, “Improving the performance of indirect evaporative cooler for energy recovery from the perspective of nozzle configuration: A CFD model analysis,” Journal of Building Engineering, vol. 76, p. 107195, 2023/10/01/ 2023.

[3] X. Ma, W. Shi, and H. Yang, “Spray parameter analysis and performance optimization of indirect evaporative cooler considering surface wettability,” Journal of Building Engineering, vol. 82, p. 108175, 2024/04/01/ 2024.

[4] X. Ma, W. Shi, Lin Lu and H. Yang, “Performance assessment and optimization of water spray strategy for indirect evaporative cooler based on artificial neural network modeling and genetic algorithm,” Applied Energy, vol. 318, 2024.

Conference papers:

[1] X. MA, W. Shi, and H. Yang. Study on water spraying distribution for improvement of indirect evaporative coolers with nozzle arrangement optimization. The 13th International Conference on Applied Energy (ICAE2021), virtual, 29 Nov - 2 Dec, 2021.

[2] X. MA, W. Shi, and H. Yang. Effect of spray nozzle parameters on surface wettability and performance improvement of indirect evaporative cooler. The 2022 MIT "A+B" Applied Energy Symposium (MITAB 2022), virtual, 5-8 July, 2022.

[3] X. MA, W. Shi, and H. Yang. Development of a wettability combined CFD model of indirect evaporative cooler with spray and water film optimization. The 14th International Conference on Applied Energy (ICAE2022), Bochum, Germany, 8-11 Aug, 2022.

中文要点｜学术成果

研究成果涵盖喷淋分布、喷嘴布置、水膜润湿性及 ANN–GA 优化，已形成期刊论文与会议论文。

Thank you!

中文要点｜致谢

感谢聆听，敬请批评指正！

Appendix

➢ Working principle

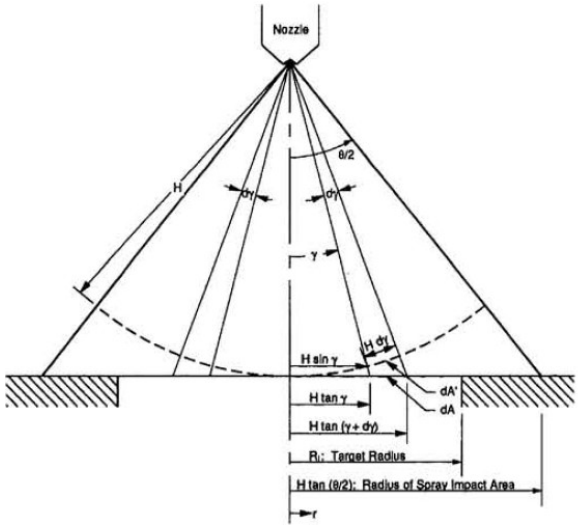


Fig. 8.2 Geometry used in volumetric flux distribution model [1]

❖ **Assumption：**

The total spray flow rate, Q, is uniformly distributed over any spherical surface centered at the spray orifice.

- **Non-Uniform Edges**: In reality, the edges of the spray area might not be perfectly straight or uniformly distributed.
- **Complex Geometries**: The actual spray might have a more complex geometry, such as an elliptical or fan-shaped distribution, which is oversimplified by a cone.

- However, the results we obtain do not necessitate fine-grained precision at every line and point, but rather within individual small grids. The foundational model has already been validated, and the model we are using has also been successfully verified. To achieve more accurate analysis, we have employed the Discrete Phase Model (DPM) in subsequent models to simulate the dispersion of spray droplets.

中文要点｜模型假设与适用性

球面均匀喷淋假设用于网格平均分析；实际喷雾存在不规则边界，后续采用 DPM 更细致地描述液滴弥散。

Appendix

➢ **Specific gaps in the four configurations**

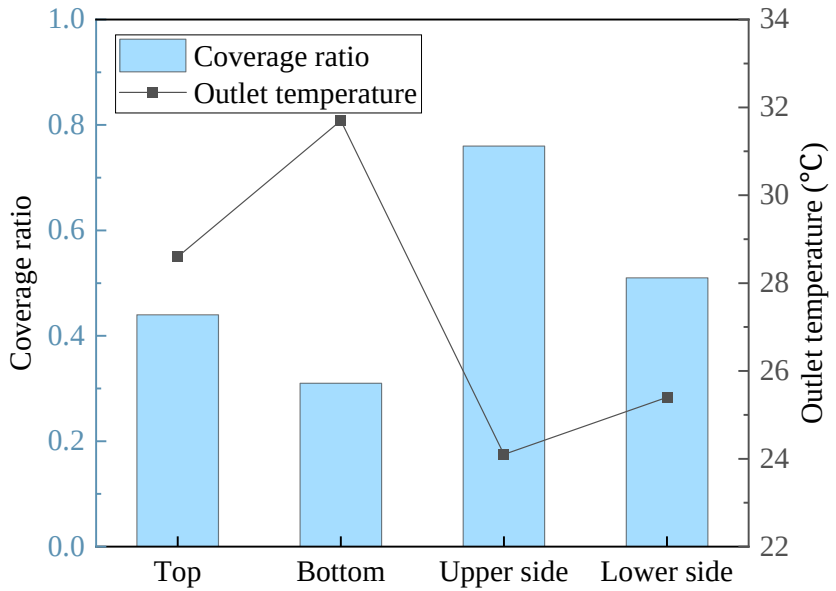


Fig. 8.5 The coverage ratio and outlet temperature of four configurations

- The **temperature drop** follows the same trend as **the water film coverage coefficient**.
- The best results are achieved in the **upper side** configuration, with a **59.2%** and **27.4%** improvement compared to the worst one.

中文要点｜四种布置的补充对比

上侧布置表现最佳；两项指标相对最差方案分别提升 **59.2%** 和 **27.4%**，体现布水方式对性能的重要影响。

Appendix

➢ IEC-ANN model

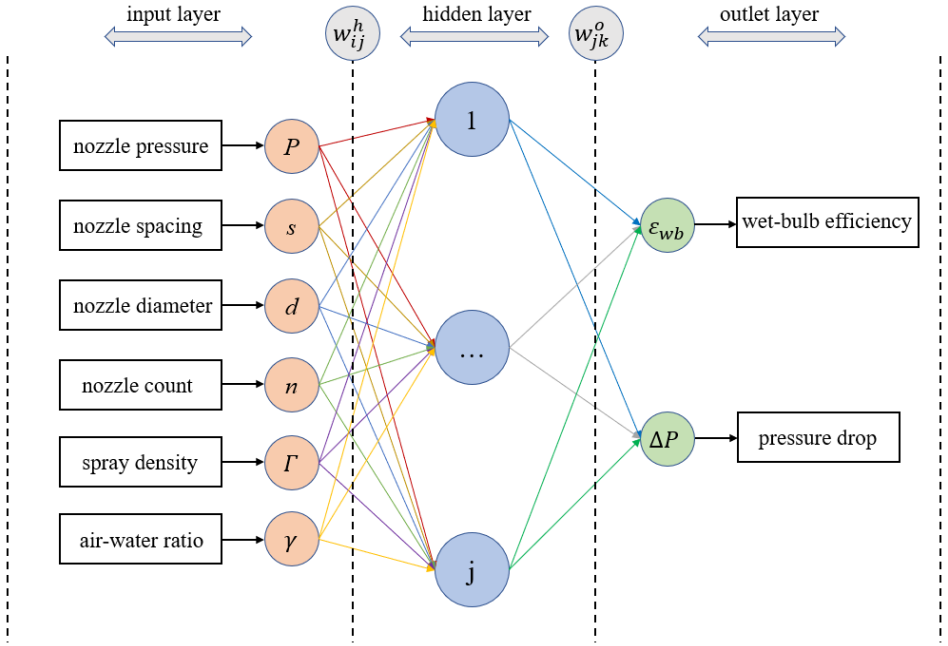


Fig.8.6 Structure of the IEC-ANN model

- **Accurate measurements:** Use precise instruments to collect data. Regularly maintain equipment to ensure consistent accuracy.
- **Replication and Validation**: Conduct multiple trials and replicate experiments to validate results.
- **Normalization and standardization**: Normalize data to ensure that different features contribute equally to the model training process.
- **Cross-Validation:** Use cross-validation techniques to assess the performance of the machine learning model. This helps in ensuring that the model generalizes well to unseen data.
- **Evaluation metrics:** The reliability of the model was verified with the evaluation parameters

中文要点｜ ANN 模型可靠性

通过精确测量、重复实验、归一化处理、交叉验证与评价指标检验，提高模型预测的可靠性和泛化能力。

Appendix

- **Validation of the film thickness**

$$\delta_w = \left(\frac{3\nu_w Re_w}{g}\right)^{1/3}$$

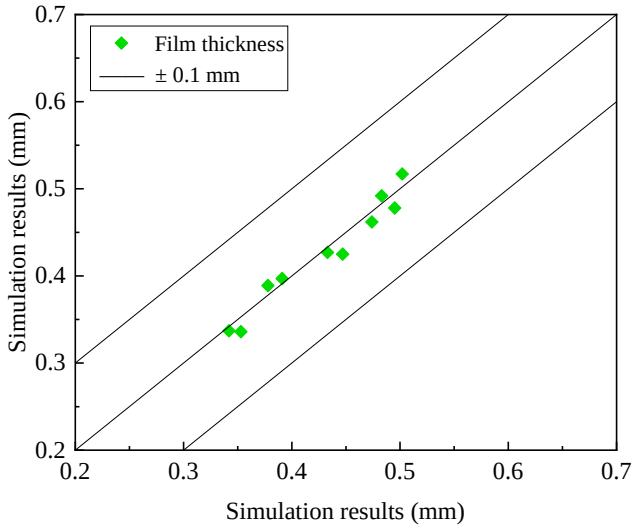


Fig. 8.7 Valdation of film thickness between simulation results and Nusselt empirical formula

- Uncertainty analysis:

$$U = f(x_1, x_2, x_3, \ldots, x_n)$$

$$\delta U = \sqrt{\left(\frac{\delta U}{\delta x_1}\delta x_1\right)^2 + \left(\frac{\delta U}{\delta x_2}\delta x_2\right)^2 + \left(\frac{\delta U}{\delta x_3}\delta x_3\right)^2 + \ldots + \left(\frac{\delta U}{\delta x_n}\delta x_n\right)^2}$$

Less than ±2.5%

中文要点｜水膜厚度验证

将数值模拟的水膜厚度与 Nusselt 经验公式对比，并开展不确定性分析。